

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LOS ANGELES REGION**

320 West 4th Street, Suite 200, Los Angeles, California 90013

ORDER NO. R4-2023-0426

FILE NO. 23-144

CI-10753

**WASTE DISCHARGE REQUIREMENTS AND
WATER RECLAMATION REQUIREMENTS
FOR
FILLMORE WATER RECYCLING PLANT**

The following Discharger is subject to the Waste Discharge Requirements (WDRs) and Water Reclamation Requirements (WRRs) set forth in this Order.

Table 1. Discharger Information

Discharger/Permittee	The City of Fillmore
Facility Name	Fillmore Water Recycling Plant
Facility Address	1580 River Street Fillmore, CA 93015 Ventura County

Table 2. Effluent Discharge Information

Discharge Locations	Fillmore WRP Percolation Ponds Former Fillmore WWTP Percolation Ponds and Trench Disposal Systems Fillmore Middle School Percolation Zones Two Rivers Park Percolation Zones
Recycled Water Use Areas	Various
Discharge Description	Disinfected tertiary recycled water
Receiving Groundwater	Pole Creek Fan Area of the Fillmore Groundwater Basin (DWR Basin No. 4-4.05)

Table 3. Administrative Information

This Order was adopted on:	December 21, 2023
This Order shall become effective on:	December 21, 2023

I, Susana Arredondo, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, Los Angeles Region, on the date indicated above.

Jenny
Newman

Digitally signed by Jenny
Newman
Date: 2024.01.10
12:09:55 -08'00'

for Susana Arredondo, Executive Officer

Adopted: December 21, 2023

The California Regional Water Quality Control Board, Los Angeles Region (Los Angeles Water Board), finds:

BACKGROUND

1. The City of Fillmore (hereafter City) provides domestic wastewater collection, treatment, and disposal services for the City of Fillmore in Ventura County. The City owns the Fillmore Water Recycling Plant (Fillmore WRP) located at 1580 River Street in Fillmore, California (Figure 1). Fillmore WRP is operated and maintained by Veolia North America under a contract with the City of Fillmore. The Fillmore WRP discharges disinfected tertiary recycled water to the soil and groundwater via percolation ponds, a trench disposal system, subsurface percolation zones, surface spray irrigation systems, and subsurface drip irrigation systems within the City of Fillmore. The City owns a separately regulated sewer collection system.
2. The Fillmore WRP was constructed and began treatment operations in September 2009 to replace the Fillmore Wastewater Treatment Plant (Fillmore WWTP) that was built in 1956. The Fillmore WWTP was subsequently decommissioned except for the percolation ponds after the completion of the Fillmore WRP.
3. Discharge from the Fillmore WRP was previously regulated under the following requirements:
 - A. Waste Discharge Requirements (WDRs) contained in Order No. R4-2006-0049 and Monitoring and Reporting Program (MRP) No. CI-1076, adopted by the Los Angeles Water Board on May 11, 2006, regulated effluent discharged to percolation ponds and effluent as recycled water for Title 22 non-potable applications.
 - B. Time Schedule Order (TSO) No. R4-2006-0050, adopted by the Los Angeles Water Board on May 11, 2006, required the City to comply with the interim effluent limits of chloride at 187 milligrams per liter (mg/L) and boron at 1.5 mg/L until the Los Angeles Water Board set limits for chloride and boron for the area of Fillmore, Santa Paula, and Piru.
 - C. Order No. R4-2017-0186, adopted by the Los Angeles Water Board on October 5, 2017, rescinded Order No. R4-2006-0049, prescribed interim effluent limits of chloride at 187 mg/L and boron at 1.5 mg/L and provided additional time to develop a regional regulatory approach to address the elevated chloride and boron concentrations in the discharge as part of the WDRs renewal for future consideration by the Los Angeles Water Board.
 - D. Order No. R4-2017-0186-A01, adopted by the Los Angeles Water Board on September 13, 2018, extended the WDRs expiration date to provide time to develop a regional regulatory approach to address elevated chloride and boron concentrations.

- E. Order No. R4-2017-0186-A02, adopted by the Los Angeles Water Board on October 8, 2020, further extended the WDRs expiration date due to the coronavirus disease 2019 pandemic (COVID-19) related relief requests and inquiries, significant loss of staff, adjustment to remote working situations, in addition to the complexity of issues, and the need to explore a long term regional salt management solution. This amendment results in the WDRs expiring on October 31, 2023, or upon issuance of new or revised WDRs for the discharge.
- 4. On November 5, 2008, the State Water Board Division of Drinking Water (DDW) conditionally approved the Title 22 Engineering Report for the Proposed Use of Recycled Water in the City of Fillmore (Title 22 Engineering Report) dated October 2008. On September 30, 2009, DDW temporarily approved the use of recycled water prior to the ultraviolet (UV) disinfection validation test. On January 20, 2010, DDW reviewed the UV validation test report dated December 2009 and subsequently provided recommendations. The final Title 22 Engineering Report was prepared by the City in November 2010.
 - A. On September 20, 2023, the City indicated that the City is in the process of obtaining DDW's acceptance for the recycled water use areas that were not included in the 2008 Title 22 Engineering Report and the area for the future expansion under the City's Recycled Water Program.
- 5. The City submitted the following information to continue discharging the treated wastewater from the Fillmore WRP.
 - A. On August 11, 2015, the City filed a Report of Waste Discharge (ROWD) with the Los Angeles Water Board for the renewal of WDRs to continue discharging treated wastewater to Fillmore WRP percolation ponds and recycled water use areas. In its 2015 ROWD, the City acknowledged the continued challenge of meeting the effluent discharge limits for chloride and boron.
 - B. On November 14, 2017, in response to Los Angeles Water Board's request for additional information on September 19, 2017, the City provided additional information, including the update on the recycled water program, an ordinance banning use of self-regenerating water softeners, a water softener buyback program, and a plan for reducing chloride and boron mass loadings to percolation ponds.
 - C. On March 22, 2018, the City proposed the "Discharge Limit Implementation" and "Strategies for Reducing Discharge Volumes of Chloride and Boron" (hereafter collectively called "effluent discharge management plan") to comply with the groundwater quality objectives of the Basin Plan for chloride and boron by limiting the mass loadings of chloride and boron to the Fillmore WRP percolation ponds. The remaining mass loadings of chloride and boron would be distributed to recycled water use areas.

- D. On February 22, 2023, the City submitted a revised ROWD to the Los Angeles Water Board. Subsequently, on March 1, 2023, the City submitted the *Discharge Impact Evaluation and Antidegradation Analysis Technical Memorandum* (Technical Memorandum) dated February 2023, part of the revised ROWD. The Technical Memorandum provides (1) the discharge impact analysis evaluation of the City's effluent discharge management plan to reduce chloride and boron mass loadings to the Fillmore WRP percolation ponds and (2) the antidegradation analysis to demonstrate no significant discharge impact to the groundwater quality.
 - E. On April 17, 2023, the City's revised ROWD was deemed complete with the submittal of supplemental information requested during Los Angeles Water Board staff's site inspection to verify the information provided in the revised ROWD on April 6, 2023.
6. Beginning in 2018, the City started discharging its effluent to the former Fillmore WWTP percolation ponds to reduce the chloride and boron loadings to the groundwater underneath the Fillmore WRP onsite percolation ponds, as shown in Table 4. During the field inspection conducted on April 17, 2023, the City stated that the effluent was routinely distributed to the former Fillmore WWTP percolation ponds and trench disposal areas. The City also routinely diverted disinfected tertiary recycled water to the subsurface percolation zones at the Two Rivers Park and Fillmore Middle School for groundwater recharge and to reduce chloride and boron loadings to the Fillmore WRP percolation ponds. Additionally, the City stated that these areas were designated as emergency percolation disposal areas, as identified in WDRs Order No. R4-2006-0049 and Order No. R4-2017-0186, as amended.

Los Angeles Water Board staff later confirmed that the City uses the subsurface percolation areas at the Two Rivers Park and Fillmore Middle School on a routine basis for disposal purposes, not for groundwater recharge or other beneficial uses. Although the City discharges the effluent to multiple locations, all disposal areas are located within the same groundwater basin. Therefore, this Order modifies the City's proposed effluent management plan by requiring the City to comply with mass-based effluent limits equivalent to the groundwater quality objectives for chloride and boron for all disposal areas, and not just at the Fillmore WRP percolation ponds.

- 7. The City provides potable water to customers in Fillmore. The source of potable water is groundwater produced from three supply wells located along the Sespe River. Many residents use self-regenerating water softeners to reduce hardness in the potable water and discharge brine generated from the water softeners into their sewer collection system.
- 8. On May 15, 2013, the Los Angeles Water Board was notified by the City that the Fillmore WRP would accept portable toilet waste from one hauler and recreational vehicles (RV) waste as authorized by the Fillmore City Council on April 23, 2013.

9. The State Water Resources Control Board (State Water Board) adopted Statewide General WDRs for Sanitary Sewer Systems (General WDRs for SSS), Order WQ 2022-0103-DWQ on December 6, 2022, which became effective on June 5, 2023. The General WDRs for SSS supersedes the previous State Water Board Order No. 2006-0003 and amendments thereafter. The General WDRs for SSS provide a consistent, statewide regulatory approach to address sanitary sewer collection systems. The General WDRs for SSS require a public, private, or other non-governmental entity that owns or operates sanitary sewer systems to develop and implement sewer system management plans and report all sanitary sewer overflows (SSOs) to the State Water Board's online SSO database. The City was enrolled under the previous General WDRs for SSS since March 18, 2009. On May 12, 2023, the State Water Board issued a Notice of Applicability for Continuation of Regulatory Coverage under the 2022 General WDRs for SSS to the City.

REGULATORY AGENCIES

10. The Los Angeles Water Board is the permitting agency for the discharge of disinfected tertiary recycled water from the Fillmore WRP. The Los Angeles Water Board issues WDRs and Water Reclamation Requirements (WRRs) to ensure that discharges of treated wastewater and recycled water from the Fillmore WRP do not adversely affect the quality of groundwater and its beneficial uses.
11. Pursuant to California Water Code (CWC) section 13523, the Los Angeles Water Board is required to consult with and receive recommendations from the DDW regarding recycled water uses to protect public health, safety, and welfare.

PURPOSE OF THIS ORDER

12. The purpose of this Order is to renew WDRs for the Fillmore WRP. This Order specifies mass-based effluent limitations based on updated ambient groundwater quality data and the results of a groundwater impact analysis consistent with State Water Board Resolution No. 68-16, *Statement of Policy with Respect to Maintaining High Quality of Waters in California* (Antidegradation Policy) and groundwater limitations based on the *Water Quality Control Plan for the Los Angeles Region: Basin Plan for the Coastal Watersheds of Los Angeles and Ventura Counties* (Basin Plan) for the protection of beneficial uses. This Order also terminates Order No. R4-2017-0186, as amended, that established interim effluent concentration limits for chloride at 187 mg/L and boron at 1.5 mg/L.
13. The WDRs are issued pursuant to Chapter 9, Division 3, Title 23, CCR and, therefore, eligible for a section 20090(a) exemption from CCR Title 27. The discharge authorized herein and the treatment and storage facilities associated with the discharge of treated municipal wastewater, except for discharges of residual sludge and solid waste, are exempt from the requirements of CCR Title 27, section 20005 et seq. The exemption, pursuant to section 20090(a) of Title 27, is based on the factors that the waste consists primarily of domestic sewage and treated effluent; that the waste discharge requirements are consistent with water quality objectives;

and that the treatment and storage facilities described herein are associated with a municipal wastewater treatment facility.

14. CWC section 13260 requires any person "proposing to discharge waste, within any region that could affect the quality of the waters of the state, other than to a community sewer system," to file a report of waste discharge. The term "waste" is defined in CWC section 13050(d) to include "sewage and any and all other waste substances, liquid, solid, gaseous, or radioactive, associated with human habitation, or of human or animal origin, ... prior to, and for purposes of, disposal." The City discharges tertiary-treated human sewage, i.e. "waste," to land where it could affect the quality of the waters of the state. Sewage contains various waste constituents, including total dissolved solids (TDS), salts (e.g., sulfate, chloride, boron), bacteria, nitrogen, priority pollutants, and constituents of emerging concern.
15. CWC section 13263(a) authorizes the Los Angeles Water Board, after any necessary hearing, to prescribe requirements as to the nature of any proposed discharge with relation to the conditions existing in the disposal area or receiving waters upon, or into which, the discharge is made or proposed. The requirements must implement any relevant water quality control plans that have been adopted, and shall take into consideration the beneficial uses to be protected, the water quality objectives reasonably required for that purpose, other waste discharges, the need to prevent nuisance, and the provisions of CWC section 13241. Under CWC section 13263(c), the requirements may include a time schedule.
16. CWC section 13267 authorizes the Los Angeles Water Board to require that any person who proposes to discharge waste submit, under penalty of perjury, technical or monitoring program reports to the Los Angeles Water Board. The estimated costs of these reports are approximately between \$25,000 and \$50,000. The burden, including costs of these reports, bears a reasonable relationship to the need for the reports and the benefits to be obtained from the reports, which include protection of public health and the environment. This Order incorporates MRP No. CI-10753 for the City (File No. 23-144), which is also necessary to ensure that the discharge of waste, and the use of recycled water, complies with this Order and is protective of human health and the environment.
17. This Order is adopted pursuant to CWC sections 13263, 13267, and 13523. It sets forth requirements, prohibitions, and other conditions to implement the Basin Plan; prescribes the limits for discharge of treated wastewater and recycled water and the City's responsibilities for the production, distribution, monitoring, and application of recycled water; and includes an MRP. The City is responsible for inspecting point-of-use facilities and ensuring compliance with the WDRs and WRRs contained in this Order.

FACILITY AND TREATMENT PROCESS DESCRIPTION

18. Fillmore WRP and Vicinity

- A. The Fillmore WRP (34°23'26" N Latitude and -118°56'26" W Longitude) is located at 1580 River Street in Fillmore [Assessor's Parcel Number (APN) is 046-0-050-140].
- B. The Fillmore WRP, its disposal areas, and recycled water use areas overlie the Fillmore Groundwater Basin-Pole Creek Fan Area (Subbasin) (California Department of Water Resource Basin No. 4-4.05), as defined in the Basin Plan (Figure 2).

19. Fillmore WRP Treatment Processes

- A. The Fillmore WRP has a treatment capacity for an average daily flowrate of 1.8 million gallons per day (mgd) and a peak flowrate of 5.0 mgd. It will have a treatment capacity for an average daily flowrate of 2.4 mgd and a peak flowrate of 6.0 mgd for future expansion.
- B. The Fillmore WRP wastewater treatment process consists of flow equalization, preliminary treatment (coarse and fine screening and grit removal), secondary anoxic and aerobic biological treatment for nitrate and biochemical chemical oxygen (BOD) removal, tertiary treatment using membrane bioreactors (MBR) for liquid/solids separation, and pathogen disinfection using ultraviolet (UV), as shown in Figure 3.

20. Fillmore WRP Effluent Quality

- A. The Fillmore WRP produces disinfected tertiary recycled water and discharges to onsite and offsite disposal areas and recycled water use areas for landscape irrigation, construction dust control, soil compaction, and demonstration wetland applications.
- B. Approximately 320 million gallons (MG) of treated water are annually discharged from the Fillmore WRP, as shown in Table 4 below.

Table 4. Annual Discharge Volume to Disposal and Recycled Water Use Areas

	Onsite WRP and Former WWTP Disposal		Offsite Disposal and Recycled Water Use		Recycled Water Use			
Year	Fillmore WRP Percolation Ponds (MG)	Former WWTP Percolation Ponds (MG)	Two Rivers Park (MG)	Fillmore Middle School (MG)	Fillmore High School (MG)	Railroad Right-of-Way (MG)	Others (MG)	Total Discharge (MG)
2013	205.6	0	20.2	64.5	8.2	25.5	0	324
2014	247.4	0	20.4	15.6	8.5	27.7	0	320
2015	254.2	0	21.5	23.3	10.2	11.5	0	321

	Onsite WRP and Former WWTP Disposal		Offsite Disposal and Recycled Water Use		Recycled Water Use			
Year	Fillmore WRP Percolation Ponds (MG)	Former WWTP Percolation Ponds (MG)	Two Rivers Park (MG)	Fillmore Middle School (MG)	Fillmore High School (MG)	Railroad Right-of-Way (MG)	Others (MG)	Total Discharge (MG)
2016	264.5	0	13.8	12.1	12.3	10.0	0	313
2017	243.6	0	10.9	36.9	10.6	7.6	0	310
2018	114.4	66.9	14.9	71.0	11.4	8.3	22.3	309
2019	35.4	167.7	3.4	72.7	5.2	3.4	16.1	311
2020	45.1	202.1	13.0	40.5	5.2	0.1	19.2	325
2021	52.0	182.6	15.4	75.1	6.9	0	16.1	348
2022	173.3	115.2	17.7	24.8	12.8	0	0.7	345
2023	0	35.9	3.2	62.3	1.3	0	0	103

Table 4 notes:

- Values in the 2023 data are for the January 2023 to March 2023 reporting period.
- The reported volumes for Two Rivers Park and Fillmore Middle School include disposal via subsurface percolation zones and recycled water landscape spray irrigation.
- The Others category includes the use of recycled water for construction dust control and soil compaction.
- The use of recycled water for landscape irrigation to the other locations, such as Riverwalk Park, Central Park, Sespe Elementary School, Fillmore Municipal Pool and Boys and Girls Club, and medians, was not included in the monitoring reports.

C. The average effluent discharge flowrate produced from the Fillmore WRP since 2013 is approximately 0.89 mgd. The average annual flowrate and effluent quality for selected constituents collected and analyzed from 2013 through March 2023 are summarized in Table 5 below.

Table 5. Average Annual Effluent Flowrate and Quality

Year	Average Flowrate (mgd)	TDS (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	Boron (mg/L)	Total nitrogen (mg/L)
2013	0.89	1,316	580	116	1.15	4.2
2014	0.88	1,345	580	112	1.08	2.4

Year	Average Flowrate (mgd)	TDS (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	Boron (mg/L)	Total nitrogen (mg/L)
2015	0.88	1,389	577	115	1.05	2.7
2016	0.85	1,408	562	126	1.06	4.6
2017	0.85	1,195	463	139	1.29	3.8
2018	0.85	1,302	512	128	1.31	3.5
2019	0.85	975	361	116	1.41	4.3
2020	0.89	1,013	397	115	1.40	6.3
2021	0.95	1,220	418	121	1.32	6.1
2022	0.94	1,198	416	125	1.46	6.5
2023	1.10	1,000	339	112	1.68	4.4

Table 5 notes:

- Values in the 2023 data are for the January 2023 to March 2023 reporting period.
- Bolded text indicates values greater than their respective groundwater quality objectives in the Basin Plan for the Fillmore Groundwater Basin-Pole Creek Fan Area.

21. Effluent Disposal and Recycled Water Applications

- A. The City operates percolation ponds, subsurface percolation zones, and a percolation trench disposal system to dispose of the treated wastewater as follows:
 - i. Two onsite percolation ponds are located within the Fillmore WRP. One is to the north, and the other is to the east of the Fillmore WRP. The Santa Clara River is located approximately 1,500 feet to the south of the Fillmore WRP. Sespe Creek is located approximately 300 feet northwest of the Fillmore WRP, as shown in Figure 4.
 - ii. Five percolation ponds are located at the former Fillmore WWTP, approximately 2,200 feet east of the Fillmore WRP and approximately 100 feet north of the Santa Clara River, as shown in Figure 5.
 - iii. One percolation trench disposal system is located at the former Fillmore WWTP and along the Santa Clara River, as shown in Figures 5 and 7.

- iv. Two subsurface percolation zones¹ are located at the Fillmore Middle School, approximately 1.2 miles northeast of the Fillmore WRP, and Two Rivers Park, north of the former Fillmore WWTP, as shown in Figures 5 and 6.
- B. Treated wastewater is disinfected and stored in the recycled water storage tank. This recycled water has residual chlorine to help prevent biofilm growth in the storage tank and distribution system. The non-potable recycled water applications are landscape irrigation, demonstration wetland, and in-plant washout, authorized by the DDW and CCR Title 22 section 60303. Use of recycled water for construction dust control and soil compaction shall be accepted by the DDW prior to implementation.
- i. The recycled wastewater is currently distributed to the following authorized areas specified in the 2008 Title 22 Engineering Report approved by DDW, as shown in blue in Figure 8 and listed below:
 - Fillmore WRP
 - Fillmore WRP Demonstration Wetland
 - Two Rivers Park
 - Riverwalk Park
 - Central Park and Central Avenue
 - Railroad Right-of-Way and "B" Street
 - Fillmore Middle School
 - Sespe Elementary School
 - Fillmore High School Baseball and Football Fields
 - Fillmore High School Sports Fields
 - Fillmore Municipal Pool and Boys and Girls Club
 - ii. During the site visit on April 17, 2023, Los Angeles Water Board staff noticed additional recycled water use areas used by the City (Figure 9) that were not included in the DDW approved 2008 Title 22 Engineering Report.
 - iii. The City's updated Recycled Water Master Plan will include expansion of the recycled water distribution system and locations to a 450-home development under construction east of Pole Creek, including approximately 5,000 feet of recycled water distribution pipeline.
 - iv. The recycled water application is feasible because utilization rate of the available assimilative capacity for chloride and boron is consistent with

¹ The City of Fillmore and its consultants called the subsurface percolation zones an "infiltrator," "effluent disposal system," "emergency disposal system," or "EDS." However, the engineering drawing provided by the City of Fillmore refers to a "subsurface percolation zone."

the Antidegradation Policy, as discussed in the antidegradation analysis section, *infra*.

HYDROGEOLOGY AND GROUNDWATER

22. The Fillmore basin is confined to the Santa Clara River Valley by the Topatopa Mountains to the north and the Oak Ridge to the south. The Fillmore Basin contains sediments that have filled the Ventura syncline. Younger alluvial sediments comprising recent sands and gravels deposited by the Santa Clara River and Sespe Creek overlie the southern and eastern portions of the basin. The Pole Creek Fan area, between Sespe Creek and the Santa Clara River, overlies the northern portion of the basin, and comprises typical alluvial fan materials.

The Fillmore Basin is considered an unconfined aquifer system. Groundwater generally flows from east to west down the axis of the basin, with southwesterly flow occurring in the Sespe Creek area. Within the San Pedro formation, there is a southerly flow component as groundwater moves from the northern part of the basin towards the valley axis, and flow continues in a westerly direction.

23. Groundwater Monitoring Network

- A. The City monitors ten groundwater monitoring wells located at the Fillmore WRP (MW-7, MW-8, MW-9, and MW-10) and in the vicinity of the former Fillmore WWTP and Two Rivers Park (MW-1, MW-2, MW-3, MW-4, MW-5, and MW-6), as shown in Figure 10. The total depths of groundwater monitoring wells range approximately from 37 to 47 feet below ground surface (bgs).
- B. The average depth to groundwater in the vicinity of the Fillmore WRP is approximately 23 feet bgs measured in 2022.
- C. The background concentrations measured from monitoring wells MW-1 and MW-2 on March 11, 1994, range from 1,200 mg/L to 1,300 mg/L for TDS, 58 mg/L to 96 mg/L for chloride, 0.8 mg/L to 0.9 mg/L for boron, and 5.2 mg/L to 8.8 mg/L for nitrate as nitrogen.
- D. The median groundwater quality at each monitoring well from 2013 through the 1st quarter of 2023 is summarized in Table 6 below.

Table 6. Median Groundwater Quality at Groundwater Monitoring Wells

Well	TDS (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	Boron (mg/L)	Nitrate as nitrogen (mg/L)	Total Coliform (MPN/100mL)
MW-1	1,270	529	109	1.12	4.95	Over 23
MW-2	1,220	472	111	1.33	5.35	2.9
MW-3	1,380	603	75	0.63	5.10	1.1

Well	TDS (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	Boron (mg/L)	Nitrate as nitrogen (mg/L)	Total Coliform (MPN/100mL)
MW-4	1,225	567	79	0.72	<0.1	<1.1
MW-5	1,250	549	95	0.92	0.05	2.2
MW-6	1,380	593	114	1.12	2.70	1.1
MW-7	1,335	584	79	0.82	4.90	2.9
MW-8	1,290	574	112	1.03	4.90	1.1
MW-9	1,295	582	105	1.00	3.80	<1.1
MW-10	1,350	581	103	1.05	3.85	1.1

Table 6 notes:

- The unit of MPN/100mL denotes the most probable number per 100 milliliters (MPN/100mL).
- Upon a separate review of fecal coliform, the median fecal coliform is less than 1.1 MPN/100mL.
- The symbol of “<” represents “less than.”
- Bolded text indicates that values are greater than their respective groundwater quality objectives in the Basin Plan for the Fillmore Groundwater Basin-Pole Creek Fan Area.

24. As stated in Finding 7, the City produces and treats groundwater to provide potable water to the residential and commercial areas in the City of Fillmore. The three groundwater supply wells are located in the Pole Creek Fan Area Subbasin of the Fillmore Groundwater Basin. Table 7 summarizes the potable water quality for the selected constituents obtained from the Annual Water Report issued by the City.

Table 7. Summary of Annual Water Reports from 2017 to 2022

Annual Water Report	TDS (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	Boron (mg/L)	Nitrate as nitrogen (mg/L) ^[1]
2017	1,080 ^[2]	455 ^[2]	95 ^[2]	1.3 ^[2]	2.1
2018	1,080 ^[2]	455 ^[2]	95 ^[2]	1.3 ^[2]	2
2019	1,007 ^[3]	445 ^[3]	69 ^[3]	1.2 ^[3]	1.8 ^[4]
2020	1,080 ^[2]	455 ^[2]	95 ^[2]	1.3 ^[2]	4.8
2021	1,190 ^[5]	486 ^[5]	73 ^[5]	1.1 ^[5]	4.9 ^[6]
2022	830 ^[7]	458 ^[8]	81 ^[8]	1.1 ^[8]	3.3

Table 7 notes:

- [1] Nitrate as nitrogen was sampled annually as the same as the reporting year of the Annual Water Report.

- [2] Sampled in 2016-2017.
- [3] Sampled in 2017-2019.
- [4] Sampled in 2018-2019.
- [5] Sampled in 2019-2020.
- [6] Sampled in 2020-2021.
- [7] Sampled in 2017, 2020-2022.
- [8] Sampled in 2020-2022.

COMPLIANCE HISTORY

25. Effluent quality data from 2013 to 2023 for the Fillmore WRP indicated that concentrations of chloride and boron were greater than their respective groundwater quality objectives in the Basin Plan for the Fillmore Groundwater Basin-Pole Creek Fan Area, as shown in Table 5 above. However, the effluent chloride and boron concentrations were below the interim effluent limits specified in TSO No. R4-2006-0050.
26. On December 19, 2013, and November 6, 2017, the Los Angeles Water Board issued Notice of Violation (NOV) letters to the City because of exceedances of chloride, boron, and total coliform concentrations in the groundwater.
27. The City responded to the NOV letters on January 16, 2014, and January 4, 2018, respectively, and stated that the exceedances were not caused by discharges from the Fillmore WRP but influenced by the natural recharge of the Fillmore Groundwater Basin from Sespe Creek.
28. The *Final Project Report for Update of Basin Plan for Piru, Sespe, and Santa Paula Hydrologic Areas* (Report) published by the California Department of Water Resources (DWR) in June 1989 indicated that the primary natural recharge of Fillmore Groundwater Basin is from percolation of the surface flows from the Santa Clara River and Sespe Creek; the subsurface inflow from Santa Felicia Basin (Lower Piru Creek); and other sources, such as percolation of surface flows from northern tributaries, surface runoff from South Mountain, precipitation, irrigation return water, and treated wastewater effluent. However, the Report stated that the Los Angeles Water Board identified discharge from the Fillmore WWTP as one of the major threats to water quality in the Fillmore Groundwater Basin.
29. As stated in Finding 7, brine wastes discharged from the residents could be one of the chloride sources. The City has issued an ordinance prohibiting the installation of new self-regenerating water softeners in 2004 to reduce influent chloride loadings to the Fillmore WRP. The City has also implemented a water softener buyback program since 2008. There were approximately 450 water softeners installed in the City of Fillmore prior to the 2004 prohibition. As of April 2023, the City has removed 99 water softeners. The City continues to promote the water softener buyback program in its water bill.

EFFLUENT DISCHARGE (MASS LOADING) MANAGEMENT

30. As stated in Finding 6, this Order modifies the City's proposal to reduce chloride and boron mass loadings by managing flowrates not only to the Fillmore WRP percolation ponds but also to all the onsite and offsite disposal areas by requiring compliance with mass-based effluent limitations. This Order allows mass loading to the Subbasin in excess of effluent limitations for recycled water application areas within the groundwater basin, as shown in Figure 11. When there is no recycled water demand, excess recycled water will be stored in an onsite recycled water storage tank (approximately 1 MG storage capacity), and chloride and boron loadings into the Subbasin will fully meet mass-based effluent limitations equivalent to the ground water quality objectives. Figure 12 illustrates the conceptual effluent management plan with a sequence of mass loading calculation.
- A. First, the total mass (Me) of each chloride and boron in the effluent will be calculated by equation (1). Second, both the mass loadings to the disposal areas (Md) in equation (2) and the mass loading to recycled water application areas (Mr) in equation (3) will be calculated. After determining the recycled water flowrate (Qr), as shown in equation (4), the total discharge flowrate (Qd) to all the disposal areas (percolation ponds in the Fillmore WRP and former Fillmore WWTP, percolation trench at the former Fillmore WWTP and subsurface percolation zones in Fillmore Middle School and Two Rivers Park) will be calculated by equation (5).
 - B. Figure 13 shows an example for determining effluent discharge flow distribution using the 2022 data listed in Table 5. Figure 13 indicates that the mass loadings of chloride or boron to the disposal areas shall not exceed 801 pounds per day (lbs/day) or 8.01 lbs/day, respectively. Under this scenario, the allowable discharge rates based on the assumed chloride and boron loadings can be calculated as 0.8 mgd and 0.241 mgd, respectively. Since boron is the limiting factor, the City can only discharge up to 0.241 mgd of the effluent to the disposal areas, including percolation ponds and subsurface percolation zones, while the remaining flowrate of 0.649 (0.89 – 0.241) mgd will need to be sent to the recycled water application areas or the 1 MG recycled water storage tank.
 - C. The February 2023 Technical Memorandum (Tech Memo) concluded that the City has greatly reduced the effluent mass loadings of chloride and boron to the onsite percolation ponds since 2018 by diverting some of the effluent for offsite disposal and recycled water use. Additionally, the Tech Memo stated that although chloride and boron effluent concentrations exceeded their respective groundwater quality objectives, the overall discharge impact to groundwater quality was minimal. Table 8 summarizes the effectiveness of managing the effluent loadings to the onsite percolation ponds.

Table 8. Effectiveness of Effluent Loading Management at the Onsite Percolation Ponds at the Fillmore WRP

Year	Effluent discharged to onsite percolation ponds (mgd)	Chloride loading to onsite percolation pond (lbs/day)	Boron loading to onsite percolation pond (lbs/day)
2016	0.72	742.8	8.23
2017	0.65	670.6	7.43
2018	0.31	319.8	3.54
2019	0.10	103.2	1.14
2020	0.12	123.8	1.37
2021	0.14	144.4	1.6

Table 8 notes:

- An average chloride concentration of 123.7 mg/L was used based on the effluent chloride concentration data collected from January 2016 to October 2022.
- An average boron concentration of 1.37 mg/L was used based on the effluent chloride concentration data collected from January 2016 to October 2022.

D. Los Angeles Water Board staff independently evaluated the City's effluent loading management practice implemented since 2018 using the data in Tables 4 and 5 above, and the results are summarized in Table 9. In order to meet the chloride and boron water quality objectives for groundwater underlying the disposal areas (onsite and former percolation ponds and offsite subsurface percolation zones), the reported discharge flowrate in Table 9 must be lower than the corresponding value in each column of chloride and boron maximum disposal flowrate in Table 9. Table 9 indicates that boron loadings to the groundwater underlying the disposal areas have met the boron groundwater quality objective, while chloride loadings have exceeded the chloride groundwater quality objective.

Table 9. Evaluation of Proposed Effluent Discharge Management to the Pole Creek Fan Area Subbasin

Year	Total effluent produced (mgd)	Maximum allowable disposal flowrate based on chloride loading (mgd)	Maximum allowable disposal flowrate based on boron loading (mgd)	Reported disposal flowrate (mgd)
2018	0.847	0.663	0.818	0.732
2019	0.852	0.733	0.817	0.765

Year	Total effluent produced (mgd)	Maximum allowable disposal flowrate based on chloride loading (mgd)	Maximum allowable disposal flowrate based on boron loading (mgd)	Reported disposal flowrate (mgd)
2020	0.888	0.770	0.854	0.822
2021	0.953	0.790	0.924	0.891
2022	0.945	0.756	0.907	0.907

Table 9 notes:

- The bolded texts indicate reported disposal flowrates greater than the calculated chloride discharge flowrate by the proposed effluent chloride loading management, suggesting that chloride (all years) was discharged in excess of the groundwater quality objective.
- Reported disposal flowrate includes flowrate to onsite percolation ponds at the Fillmore WRP, former percolation ponds at the former Fillmore WWTP, offsite subsurface percolation zones and landscape irrigation at the Two Rivers Park, and offsite subsurface percolation zones and landscape irrigation at the Fillmore Middle School. Note that monitoring reports submitted by the City did not report the separate flowrates for the subsurface percolation zones and landscape irrigation at the Two Rivers Park and Fillmore Middle School.

E. The City's proposed Effluent Discharge Management Plan is feasible because the City has constructed recycled water infrastructures to divert the effluent to recycled water use areas, and the City has a 1-million gallon recycled water storage tank onsite to control the flow to the disposal areas and recycled water use areas. Additionally, this Order requires the City to comply with effluent mass loading to the disposal areas for both chloride and boron.

31. The City currently manages effluent discharge to the percolation ponds and subsurface percolation zones and continues to identify additional recycled water users to increase beneficial reuse of treated wastewater. When there is no recycled water demand, all effluent discharges to the disposal areas will need to meet groundwater quality objectives for chloride and boron of the Basin Plan.

PER- AND POLYFLUOROALKYL SUBSTANCES INVESTIGATION

32. Per- and polyfluoroalkyl substances (PFAS) are a group of more than 12,000 human-made substances that are not naturally occurring and are resistant to heat, water, and oil. PFAS have been widely used as surface coatings and protectant formulations in consumer goods, such as carpet and home textiles, clothing, food packaging, and non-stick cookware. PFAS have also been used as a surfactant in chrome plating, firefighting foam, and other industrial applications. PFAS are resistant to degradation and do not break down in the environment. They can accumulate within the human body and are toxic at relatively low concentrations.

33. On July 9, 2020, the State Water Board issued Order WQ 2020-0015-DWQ requiring certain publicly owned treatment works (POTWs) including Fillmore WRP to investigate PFAS in the influent, effluent, groundwater, and biosolids. The City has completed the investigation as required by Order WQ 2020-0015-DWQ, and the results are summarized below. This Order now requires the City to monitor the constituents of emerging concern (CEC), including perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS).

No PFAS were detected in biosolids that were sampled on December 8, 2020. However, perfluorobutanoic acid (PFBA) and perfluoropentanoic acid (PFPeA) at concentrations of 42 nanograms per liter (ng/L) and 74 ng/L, respectively were found in the influent. In addition, PFAS were also found in the effluent and groundwater samples, as shown in Tables 10 and 11 below.

Table 10. Analytical Results for PFAS Detected in Effluent

Sampled Date	PFBA (ng/L)	PFPeA (ng/L)	PFHxA (ng/L)	PFOA (ng/L)	PFBS (ng/L)
12/8/2020	ND	60	39	ND	ND
2/17/2021	7.4	81	83	13	ND
5/11/2021	8.3	62	37	12	ND
8/17/2021	13	49	81	9.7	15

Table 10 notes:

- ND denotes not detected.
- PFHxA denotes perfluorohexanoic acid.
- PFBS denotes perfluorobutanesulfonic acid.

Table 11. Analytical Results for PFAS Detected in Groundwater

Constituent	Unit	MW-7	MW-8	MW-9	MW-10
PFBS	ng/L	7.9	57	8.2	ND
PFHxS	ng/L	7.5	12	4.6	ND
PFBA	ng/L	ND	8	ND	ND
PFPeA	ng/L	ND	79	ND	ND
PFHxA	ng/L	ND	62	ND	ND
PFOA	ng/L	ND	16	ND	ND
PFOS	ng/L	ND	4.6	ND	ND

Table 11 notes:

- PFHxS denotes perfluorohexanesulfonic acid.

34. The DDW established notification levels (NLs) and response levels (RLs) for several PFAS, which are nonregulatory, health-based advisory levels for drinking water based on the California Office of Environmental Health Hazard Assessment recommendations, as shown in Table 12. DDW will be developing maximum contaminant levels (MCLs) for PFOA and PFOS drinking water standards. Additionally, the State Water Board and California Environmental Protection Agency (CalEPA) have been collaborating with the United States Environmental Protection Agency (USEPA) and other governmental agencies over PFAS concerns since 2012.

Table 12. Notification and Response Levels for Drinking Water

Constituent	Notification Level (ng/L or ppt)	Response Level (ng/L or ppt)	Date Established
PFOA	5.1	10	February 6, 2020
PFOS	6.5	40	February 6, 2020
PFBS	500	5,000	March 5, 2021
PFHxS	3	20	October 31, 2022

Table 12 note:

- The unit of ppt denotes parts per trillion.

APPLICABLE PLANS, POLICIES, AND REGULATIONS

35. ***Water Quality Control Plan for the Coastal Watersheds of Los Angeles and Ventura Counties (Basin Plan)*** – The Basin Plan is designed to preserve and enhance water quality and protect the beneficial uses of all regional waters. Specifically, the Basin Plan (i) designates beneficial uses for surface and ground waters, (ii) establishes narrative and numeric water quality objectives that must be attained or maintained to protect the designated beneficial uses, and (iii) sets forth implementation programs to protect the beneficial uses of the waters of the state. The Basin Plan also incorporates State Water Board Resolution No. 68-16, *Statement of Policy with Respect to Maintaining High Quality of Waters in California* (“Antidegradation Policy”). In addition, the Basin Plan incorporates by reference applicable State and Los Angeles Water Board plans and policies and other pertinent water quality policies and regulations. Since 1994, numerous Basin Plan amendments have been adopted and more current background, program, and geographical information have become available. This Order implements the plans, policies and provisions of the Los Angeles Water Board’s Basin Plan.
36. The Basin Plan (Chapter 3) establishes water quality objectives for groundwater. The chemical constituents and radioactivity water quality objectives for groundwater designated as domestic or municipal supply (MUN) incorporate the primary and secondary maximum contaminants levels (MCLs) for inorganic, organic, and radioactive contaminants in drinking water that are codified in CCR Title 22, Division

1 (CCR Title 22). This incorporation by reference is prospective, including future changes to the incorporated provisions as the changes take effect. Also, the Basin Plan establishes the following taste and odor water quality objective for groundwater: “Ground waters shall not contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.” These water quality objectives are implemented in this Order to protect groundwater quality.

37. The Fillmore Groundwater Basin-Pole Creek Fan Area (Subbasin) is in a triangular area east of Sespe Creek and north of the Santa Clara River. The City of Fillmore and the Fillmore WRP, including disposal and recycled water use areas are located on this Subbasin. The applicable groundwater quality objectives for the Subbasin are summarized in Table 13 below.

Table 13. Groundwater Quality Objectives for the Fillmore Groundwater Basin-Pole Creek Fan Area (Subbasin)

TDS (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	Boron (mg/L)	Nitrogen (mg/L)	Coliform (MPN/100mL)
2,000	800	100	1.0	10	1.1

Table 13 notes:

- Nitrogen refers to the sum of the nitrate as nitrogen and nitrite as nitrogen.
- Coliform refers to either total coliform or fecal coliform.

38. In addition, the Basin Plan incorporates State Water Board Resolution No. 88-63, which establishes state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. Beneficial uses applicable to the receiving water are shown in Table 14 below.

Table 14. Basin Plan Beneficial Uses of Groundwater

Receiving Water	Beneficial Uses
Fillmore Groundwater Basin-Pole Creek Fan Area (DWR Basin No. 4-4.05)	Municipal and domestic supply (MUN), industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR).

39. **CCR Title 22** – CCR Title 22 contains primary and secondary MCLs for inorganic, organic, and radioactive contaminants in drinking water. These MCLs are codified in CCR Title 22. Title 22 primary MCLs have been incorporated into the Basin Plan as water quality objectives (see Attachments B-1 to B-6). MCLs are used as one of the primary bases for effluent limits for discharges of recycled water in WDRs and WRRs to protect the designated beneficial uses of municipal and domestic supply.
40. **Publicly Owned Treatment Works (POTW)** – The term POTW means a treatment works as defined by section 212 of the federal Clean Water Act, which is owned by

a State or municipality (as defined by section 502(4) of the Act). This definition includes any devices and systems used in the storage, treatment, recycling and reclamation of municipal sewage or industrial wastes of a liquid nature. It also includes sewers, pipes and other conveyances only if they convey wastewater to a POTW treatment facility. The term also means the municipality as defined in section 502(4) of the Clean Water Act, which has jurisdiction over the indirect discharges to and the discharges from such treatment works. (40 Code of Federal Regulations [CFR] 403.3(q)). Fillmore WRP meets all above criteria and therefore is considered a POTW.

41. **Assembly Bill No. 685 (AB 685) – CWC section 106.3** – It is the policy of the State of California that every human being has the right to safe, clean, affordable, and accessible water adequate for human consumption, cooking, and sanitary purposes. This Order promotes that policy by requiring discharges to meet maximum contaminant levels developed to protect human health and ensure that water is safe for domestic use.
42. CWC section 13263 authorizes the Los Angeles Water Board, after any necessary hearing, to prescribe requirements as to the nature of any proposed discharge with relation to the conditions existing in the disposal area or receiving waters upon, or into which, the discharge is made or proposed. The requirements must implement any relevant water quality control plans that have been adopted, and shall take into consideration the beneficial uses to be protected, the water quality objectives reasonably required for that purpose, other waste discharges, the need to prevent nuisance, and the provisions of CWC section 13241.

A. Past, present, and probable future beneficial uses of water;

The receiving water for discharges from the Fillmore WRP is the Fillmore Groundwater Basin-Pole Creek Fan Area Subbasin. The receiving water limitations in this Order are specified to maintain the beneficial uses of this Subbasin: municipal and domestic water supply (MUN), industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR).

As stated in Finding 5.A and the Compliance History section above, the City acknowledged the challenge to comply with the chloride and boron effluent limits. To address the chloride and boron discharge issues, the City implemented its effluent discharge management plan. The results of the antidegradation analysis indicate that the effluent discharge management allows beneficial use of disinfected tertiary recycled water with minimal impact to groundwater quality. Since Fillmore WRP produces disinfected tertiary recycled water, it is available to the community in the City of Fillmore for agricultural supply and additional landscape irrigation.

Since this Order specifies effluent and groundwater limitations and includes effluent and receiving water monitoring and reporting requirements to verify that discharges will not adversely affect the beneficial uses of groundwater now

or in the future, it protects and maintains past, present, and probable future beneficial uses of water.

- B. Environmental characteristics of the hydrographic unit under consideration, including the quality of the water available thereto;

This Order implements the site-specific water quality objectives for the Fillmore Groundwater Basin-Pole Creek Fan Area Subbasin in the Basin Plan, considering geology, hydrogeology, and hydrology. Based on recent and historical data, the regional groundwater basin currently has high-quality water, but is experiencing increases in salt and nitrogen loading from natural and anthropogenic sources. The Fillmore WRP produces effluent quality that is poorer than the groundwater quality objectives, but the implementation of the City's effluent discharge management plan will comply with the state's Antidegradation Policy (Resolution No. 68-16) and allow use of the groundwater basin's assimilative capacity for recycled water applications consistent with the maximum benefit to the people of the State.

- C. Water quality conditions that could reasonably be achieved through the coordinated control of all factors that affect water quality in the area;

The water quality conditions that can be reasonably achieved are those consistent with the site-specific water quality objectives necessary to protect the beneficial uses of the groundwater basin through the coordinated control of all factors that affect water quality in the area. The Fillmore WRP produces disinfected tertiary recycled water, which ensures groundwater quality is not impacted with nutrients, such as nitrogen and organic compounds, but does not remove salts from the effluent. To achieve the groundwater quality objectives for chloride and boron, the City has implemented (1) an effluent discharge management plan to reduce chloride and boron loadings to groundwater when there is no recycled water demand, (2) a self-regenerating water softener buyback program, (3) recycled water applications, and (4) an ordinance to prohibit new water softener installation. Additionally, this Order requires the City to meet the mass-based effluent limits for chloride and boron discharges to disposal areas at all times. Consequently, these limitations will manage flowrate to the Subbasin and reduce the chloride and boron loadings to the Subbasin when there is no demand for recycled water applications. In the future, the Los Angeles Water Board may separately consider whether to require the City to clean up underlying polluted groundwater. The water quality objectives implemented in this Order are reasonably achievable through the City's continued performance of the programs listed above.

- D. Economic considerations;

The Legislature did not define "economic considerations" in CWC section 13241. In *City of Duarte v. State Water Resources Control Board et al.*, the Court of Appeal held that "...the Water Control Boards are charged with

taking into account economic considerations, not merely costs of compliance with a permit ... economic considerations also include, among other things, the costs of not addressing the problems of contaminated water." (*City of Duarte*, 60 Cal.App.5th 258, 276 (considering 13241 factors in context of municipal separate storm sewer system or MS4 permit).) Indeed, the "manner in which the Water Control Boards consider and comply with Water Code section 13241 is within their discretion." (*Id.*, at p. 273, citing *City of Arcadia v. State Water Resources Control Board* (2006) 135 Cal.App.4th 1392, 1415.) Since the Los Angeles Water Board has broad discretion in how it considers this factor, the Board interprets this factor as not only requiring a consideration of the costs of compliance, but also other relevant economic factors such as the societal and environmental costs of not adequately controlling discharges.

The City of Fillmore is not a disadvantaged community (DAC) as defined in CWC section 79505.5(a), which provides, "disadvantaged community' means a community with an annual median household income that is less than 80 percent of the statewide annual median household income." The United States Census reported that the 2021 California median house income is \$81,575, and the City of Fillmore's median household income is \$78,024. However, the City of Fillmore is partially a DAC based on the 2021 median household income data at the block group level.

An upgrade of the Fillmore WRP to remove salts from the effluent would require installation of a desalination treatment system, such as a reverse osmosis (RO) system. An RO system would require a capital cost investment of approximately \$13 million and annual operations and maintenance (O&M) cost of \$1.6 million, which will have a significant economic impact on the community. The City's sewer service rates has been \$103.36 per unit since 2020, one of the highest sewer service rates in California. With implementation of the effluent discharge management plan, the effluent from the Fillmore WRP could minimize adverse impacts to groundwater quality, while attaining beneficial uses and water quality objectives and complying with the Antidegradation Policy.

This Order requires the City to implement a compliance schedule to comply with the mass loading effluent limitations for chloride and boron by conducting a feasibility study to determine the most appropriate approach to compliance. This feasibility study and selected methodology implementation to the Fillmore WRP may impact ratepayers, but will help protect public health, the environment, and the beneficial uses in the long term.

E. The need for developing housing within the region;

The United States Census reported that the population of the City of Fillmore was 16,419 in 2020. As stated in Finding 21.B.iii, residential development is under construction and will help address the longstanding housing shortage in California. The Fillmore WRP is designed to treat up to 5.0 mgd of wastewater

and is capable of future expansion to 6.0 mgd. The Fillmore WRP currently discharges approximately 0.96 mgd of treated wastewater in 2022. Therefore, the Fillmore WRP has sufficient capacity to treat additional wastewater generated from newly developed residential, commercial, and industrial districts.

F. The need to develop and use recycled water;

This Order authorizes the City to treat and discharge up to 5.0 mgd of disinfected tertiary recycled water to generate recycled water for landscape irrigation pursuant to CCR Title 22 section 60304(a). This will help offset the need for potable water.

43. In accordance with CWC section 13263(g), no discharge of waste into waters of the state, whether or not the discharge is made pursuant to waste discharge requirements, shall create a vested right to continue the discharge. All discharges of waste into waters of the state are privileges, not rights.
44. **State Water Board Resolution No. 77-1** – The State Water Board adopted Resolution No. 77-1, *Policy with Respect to Water Recycling in California*, which includes principles that encourage and recommend funding for water recycling and its use in water-short areas of the State.
45. **Recycled Water Policy** – State Water Board Resolution No. 2009-0011, *Adoption of a Policy for Water Quality Control for Recycled Water* (Recycled Water Policy) as amended in 2013 and 2018, is intended to support the State Water Board’s Strategic Plan to promote sustainable local water supplies. Increasing the acceptance and promoting the use of recycled water is a means towards achieving sustainable local water supplies and can result in reduction in greenhouse gases, a significant driver of climate change. The Recycled Water Policy is also intended to encourage beneficial use of, rather than solely disposal of, recycled water generated from municipal wastewater sources in a manner that fully implements state and federal water quality laws.

The most recent amendment to the Recycled Water Policy was adopted on December 11, 2018 and became effective on April 8, 2019. The amended Recycled Water Policy requires wastewater and recycled water dischargers to annually report monthly volumes of influent, wastewater produced, and effluent, including treatment level and discharge type. As applicable, dischargers are additionally required to annually report recycled water use by volume and category of reuse. This Order implements the reporting requirements of the Recycled Water Policy.

The Recycled Water Policy requires the development of salt and nutrient management plans (SNMPs) in basins where salts and/or nutrients are a threat to water quality, and therefore, planning is needed to achieve water quality objectives in the long term. SNMPs are required to include a monitoring program, source identification, assimilative capacity estimation, implementation measures including

an antidegradation analysis to manage or reduce salt and nutrient loading in the basin on a sustainable basis.

The Los Angeles Water Board amended the Basin Plan to incorporate management measures in the Lower Santa Clara River SNMP dated June 2015 under the title, "Groundwater Quality Management Measures for Salt and Nutrients in the Lower Santa Clara River Basin," Resolution No. R15-007, on July 9, 2015. The Office of Administrative Law approved the Basin Plan amendment on June 19, 2018.

46. **Antidegradation Policy, State Water Board Resolution No. 68-16** – State Water Board Resolution No. 68-16, *Statement of Policy with Respect to Maintaining High Quality of Waters in California* (Antidegradation Policy) requires the Los Angeles Water Board, in regulating the discharge of waste, to maintain high-quality waters of the state until it is demonstrated that any change in quality will be consistent with maximum benefit to the people of the State, will not unreasonably affect beneficial uses, and will not result in water quality less than that described in the State Water Board's policies (e.g., quality that exceeds water quality objectives). The Los Angeles Water Board finds that the discharge, as allowed in these WDRs, is consistent with Resolution No. 68-16 since this Order requires (1) compliance with the requirements set forth in this Order, including the use of best practicable treatment and control (BPTC) technology, (2) implementation of the revised MRP; and (3) regular monitoring of the discharge to comply with water quality objectives.

ANTIDEGRADATION ANALYSIS

47. The antidegradation analysis in the Tech Memo was conducted by using a mixing mass balance spreadsheet model to evaluate the discharge impact to groundwater underlying the percolation ponds at the Fillmore WRP. The Tech Memo indicated that two scenarios were established to evaluate groundwater impact for an extended period and distance from the percolation ponds, as shown in Figure 14. Scenario 1 estimates the impact of effluent that is discharged to the percolation ponds at the Fillmore WRP, which is approximately 43% of the total effluent, while Scenario 2 estimates the impact to groundwater if all effluent is discharged to the percolation ponds at the Fillmore WRP. In other words, Scenario 1 represents operation of effluent discharge management with approximately 57% of the recycled water implementation, and Scenario 2 represents an operation without effluent discharge management.
48. Table 15 shows the average groundwater quality and available assimilative capacity of TDS, chloride, and nitrate as nitrogen in the Subbasin. The data were taken from the 2015 SNMP for Lower Santa Clara River Basin, including the Subbasin. The groundwater quality data were collected from 1996 through 2012. The available assimilative capacity for the Subbasin is defined as the difference between the groundwater quality objectives of the Basin Plan and the ambient groundwater quality.

Table 15. Ambient Groundwater Quality and Available Assimilative Capacity for the Subbasin in the 2015 SNMP

Parameter	TDS (mg/L)	Chloride (mg/L)	Nitrate as nitrogen (mg/L)
Groundwater Quality Objectives	2,000	100	10
Ambient Groundwater Quality	1,101	59	2.9
Available Assimilative Capacity	899	41	7.1

49. Since the 2015 SNMP does not include the boron assimilative capacity data for the Subbasin, the City independently compiled groundwater quality data from its groundwater monitoring wells, municipal water wells, domestic water wells, and agricultural supply wells within the Subbasin to determine the available assimilative capacity for boron. In the same manner, the site-specific ambient groundwater quality and available assimilative capacity for chloride were determined.
50. Table 16 lists general water quality in the vicinity of the Fillmore WRP used by the City for conducting a site-specific antidegradation analysis. Based on the site-specific groundwater quality data collected from 2016 to 2022 at monitoring wells in the Subbasin, the available assimilative capacity for chloride and boron are 14 mg/L and 0.12 mg/L, respectively.

Table 16. General Water Quality Used for the Site-specific Antidegradation Analysis

Parameter	Chloride (mg/L)	Boron (mg/L)
Groundwater Quality Objective	100	1.0
Ambient Groundwater Quality	86	0.88
Available Assimilative Capacity	14	0.12
Groundwater Inflow Quality	75	0.70
95 th Percentile Effluent Quality	144	1.71

51. The groundwater mixing model considers two major contributions of chloride and boron to the Subbasin: (1) regional groundwater inflow and (2) effluent discharge from the Fillmore WRP. Concentrations of chloride and boron in the regional groundwater inflow were determined using the groundwater quality from the upgradient monitoring wells. The average and 95th percentile effluent concentrations of chloride and boron were used to evaluate the discharge impact to groundwater.
52. The Tech Memo predicts groundwater concentrations at the distance of 1,500, 2,000, 2,500, 3,000, and 3,500 feet from the Fillmore WRP percolation ponds over a period of 1, 5, 10, 20, and 30 years using average and 95th percentile effluent

concentrations under the following two scenarios: Scenario 1 represents implementation of City's effluent discharge management plan that includes onsite and offsite disposal areas and uses of recycled water, and Scenario 2 does not implement an effluent discharge management plan and assumes disposal of effluent to onsite percolation ponds only. Tables 17 and 18 summarize the result of the groundwater impact analysis.

Table 17. Estimated Time Required in Years for Groundwater Quality to Exceed Groundwater Quality Objectives for Chloride and Boron at Different Distances from the Percolation Ponds at Average Effluent Concentrations

Constituent	Scenario	1,500 feet	2,000 feet	2,500 feet	3,000 feet	3,500 feet
Chloride	1	Over 30	Over 30	Over 30	Over 30	Over 30
Chloride	2	2	3	5	10	20
Boron	1	Over 30	Over 30	Over 30	Over 30	Over 30
Boron	2	2	10	Over 30	Over 30	Over 30

Table 17 notes:

- Scenario 1 represents implementation of City's effluent discharge management plan with approximately 57% of the recycled water implementation.
- Scenario 2 does not implement an effluent discharge management plan and assumes disposal into onsite percolation ponds only.

Table 18. Estimated Time Required in Years for Groundwater Quality to Exceed Groundwater Quality Objectives for Chloride and Boron at Different Distances from the Percolation Ponds at 95th Percentile Effluent Concentrations

Constituent	Scenario	1,500 feet	2,000 feet	2,500 feet	3,000 feet	3,500 feet
Chloride	1	4	20	Over 30	Over 30	Over 30
Chloride	2	1	1	2	3	4
Boron	1	2	10	Over 30	Over 30	Over 30
Boron	2	1	1	2	4	9

53. In summary, this Order requires the City to manage chloride and boron mass loadings to the onsite and offsite disposal areas to ensure that mass-based effluent limits are met by managing their mass loading rates. However, the plan allows chloride and boron loadings to the Subbasin in excess of the mass-based effluent limits when there is demand for recycled water applications. Based on the average effluent concentration and flowrate, the City's effluent discharge management plan (Scenario 1 in Table 17) will not cause exceedance of the groundwater quality

objectives for chloride and boron for at least 30 years when implementing the recycled water applications up to 53%. However, when discharging the 95th percentile effluent concentration and flowrate are used as the worst case scenario, the available assimilative capacity for boron in the Subbasin will be depleted within a couple of years, as shown in Table 18.

Therefore, the MRP requires the City to annually report the ambient groundwater quality of the Subbasin to monitor the available assimilative capacity of the Subbasin. This Order also requires the City to submit a workplan to prevent further groundwater degradation when 10 percent of the Subbasin assimilative capacity for chloride and boron is used up.

Based on the Tech Memo, the effluent discharge management plan will utilize less than 10 percent of the Subbasin assimilative capacity for chloride and boron and will not produce a significant reduction of water quality. As a result, beneficial uses of the Subbasin will be maintained.

54. The Antidegradation Policy requires that disposal of waste into the waters of the state be regulated to achieve the highest water quality consistent with the maximum benefit to the people of the state. High quality waters shall be maintained to the maximum extent possible consistent with the Antidegradation Policy. This Order establishes mass-based effluent limitations consistent with Resolution No. 68-16 and the State Water Board's Recycled Water Policy and receiving water limitations consistent with the Basin Plan. Additionally, this Order contains requirements for maintaining the best practicable technology and control (BPTC) and the high-water quality consistent with the maximum benefit to the people of the state by implementation of the revised MRP. Therefore, this Order protects all beneficial uses specified in the Basin Plan.

GLOBAL WARMING AND CLIMATE CHANGE

55. In Southern California, the predicted impacts of climate change are numerous, including the following.

Annual average temperatures are expected to increase, coupled with a higher frequency of extreme heat days. A likely consequence of this warmer climate will be more severe drought periods, leading to an increase in the amount and intensity of fires and a longer fire season. In addition, precipitation patterns are likely to be modified.

A decrease in snowfall, combined with warmer temperatures, will induce a decrease in the amount and duration of snowpack, an essential source of freshwater to the region. Although changes to mean precipitation are expected to be small, the increasing occurrence of extreme precipitation events will amplify the risk of flooding. These impacts may affect water quality in multiple ways, including decreases in stream flow, reductions in, and changes to, aquatic habitats, increases in surface

water temperature, increases in pollutant levels, sedimentation, algal growth, and changes in salinity levels and acidification in coastal areas.

For permitted facilities such as POTWs, specific impacts could include, but not limited to, an increase in the concentration of pollutants entering the facility, an increase in the temperature of effluents and receiving waters, an increase in storm water inflow and infiltration, increase in flooding inundation of facilities, sewer overflows, power outages, pump maintenance issues, and onsite or nearby hillside destabilization.

56. On March 7, 2017, the State Water Board adopted Resolution No. 2017-0012, *Comprehensive Response to Climate Change*, recognizing the challenges posed by climate change, directed state agencies to take climate change into account in their planning decisions, guided by the following principles: Priority should be given to actions that both build climate preparedness and reduce greenhouse gas emissions; where possible, flexible and adaptive approaches should be taken to prepare for uncertain climate impacts; actions should protect the state's most vulnerable populations; and natural infrastructure solutions should be prioritized.
57. On May 10, 2018, the Los Angeles Water Board adopted Resolution No. R18-004, A Resolution to Prioritize Actions to Adapt to and Mitigate the Impacts of Climate Change on the Los Angeles Region's Water Resources and Associated Beneficial Uses, which encourages mitigating direct and indirect impacts of climate change on water quality and beneficial uses.
58. The Fillmore WRP is located outside of the 100-year floodplain, and hence, no flooding issues are expected. The majority of the Fillmore WRP site, including disposal and recycled water use areas, are identified as areas with reduced flood risk due to presence of a levee. Although some of the disposal and recycled water use areas along the Santa Clara River are located within the 100-year floodplain, they have not been significantly impacted by flood events since the Fillmore WRP was constructed.
59. This Order contains provisions to require planning and taking actions to address climate-related impacts that can cause or contribute to violations of permit requirements and/or degradation of waters of the state (see, Section E.11, *infra*).

RATIONALE FOR LIMITATIONS

60. The numeric groundwater limitations imposed by this Order are based on the following.
 - A. TDS, sulfate, chloride, and boron – Groundwater limitations are based on the groundwater quality objectives set forth for the Subbasin in the Basin Plan and as shown in Table 13.

- B. Nitrogen compounds and bacteria parameters – Groundwater limitations are based on the regional groundwater objectives set forth by the Basin Plan and as shown in Table 13.
61. The numeric effluent limitations imposed by this Order are based on the following.
- A. Biochemical oxygen demand (BOD) and total suspended solids (TSS) – Effluent limitations are technology-based limits contained in similar orders for POTWs indicative of treatment levels that are achievable by tertiary-treated wastewater treatment systems.
 - B. Oil and grease – Effluent limitations are based on the narrative water quality objective for surface waters in the Basin Plan and on best professional judgment to interpret the narrative objective. The same numeric limitations are specified in other permits for tertiary-treated wastewater treatment plants adopted by this Los Angeles Water Board.
 - C. Total dissolved solids (TDS), sulfate, nitrogen compounds, fecal coliform, and enterococcus – Effluent limitations are based on the Basin Plan groundwater quality objectives.
 - D. Methylene blue activated substances (MBAS) – Effluent limitations are based on the water quality objective in the Basin Plan consistent with CCR Title 22 secondary MCLs.
 - E. Turbidity and total coliform – Effluent limitations are technology-based requirements specified in CCR Title 22 for recycled water applications.
 - F. Chloride and boron – Effluent limitations are mass-based limits to ensure attainment of groundwater limitations by implementation of the effluent discharge management plan.

RATIONALE FOR COMPLIANCE SCHEDULE AND MODIFICATION OF AVERAGING PERIODS FOR CHLORIDE AND BORON

62. On September 20, 2023, the City notified the Los Angeles Water Board that the effluent cannot meet the weekly mass-based effluent limitations for chloride and boron specified in the tentative WDRs and WRRs because of the operational challenges for the flow management and the storage shortage when there is low demand for recycled water during the rainy season. The City discovered the storage shortage because the existing recycled water storage tank is designed to provide hydraulic pressure for recycled water distribution and not for excess recycled water storage purposes. Although the Fillmore WRP has a design capacity of 5 mgd, the City stated that the Fillmore WRP is unable to hold excess recycled water. Currently, the City does not have alternative measures to divert excess treated wastewater, exceeding the mass loading equivalent to the groundwater quality objectives or exceeding agronomic rate at the landscape irrigation areas, to other locations.

Therefore, the City needs to develop a plan to comply with the final effluent limitations for chloride and boron, as specified in Section B of this Order, *infra*.

Additionally, this Order requires the City to comply with mass-based final effluent limitations based on a 3-month rolling average. The 3-month rolling average compliance averaging period is consistent with the Literature Review Evaluation supplemental study for the Upper Santa Clara River Chloride Total Maximum Daily Load (TMDL) (Newfields, 2008) for agricultural beneficial uses for salt-sensitive crops.

63. In order to comply with the effluent limitations for chloride and boron (Sections B.6 and B.7), interim effluent limitations are necessary until installing an additional storage tank, expanding recycled water applications, investigating better water supply sources, improving treatment processes (e.g., reverse osmosis, zero liquid discharge for brine disposal, etc.), or another effective method of compliance to be identified by the City, as required by Provisions I.21, *infra*, are completed. The interim concentration-based effluent limitations for chloride and boron are established by a 95th percentile of the 12-month rolling average from the 3rd quarter 2017 through the 2nd quarter 2023 effluent data reported and specified in Table 20 below.

CALIFORNIA ENVIRONMENTAL QUALITY ACT AND PUBLIC NOTIFICATION

64. In accordance with the provisions of the California Environmental Quality Act (CEQA) (Public Resources Code section 21000 et seq.), the City prepared and circulated an Environmental Impact Report for public comments. On May 24, 2005, the City of Fillmore Council adopted City Council Resolution No. 05-2817, certifying the adequacy of the final Environmental Impact Report (EIR) for the Fillmore WRP project.
65. The Los Angeles Water Board, a responsible agency for the purpose of CEQA, reviewed and considered the information contained in the final EIR in drafting this Order, and finds that the final EIR sufficiently addresses and mitigates any environmental impacts.
66. The Los Angeles Water Board conducted a community profile assessment based on CalEnviroScreen 4.0 for the Census Tract Nos. 6111000302, 6111000303, and 6111000304 defined by CalEnviroScreen. CalEnviroScreen is a mapping tool that helps identify California communities that are most affected by many sources of pollution, and where people are often especially vulnerable to pollution's effects. It uses environmental, health, and socioeconomic information to produce scores for every census tract in the state. The scores are mapped so that different communities can be compared. An area with a high score is one that experiences a much higher pollution burden than areas with low scores. CalEnviroScreen ranks communities based on data that are available from state and federal governmental sources.

For the community profile which comprises racial demographics, pollution burdens and income survey data, the Los Angeles Water Board found that (1) this Order may disproportionately impact Black, Indigenous, people of color (BIPOC) communities due to the community made up of 24% white, 76% BIPOC (0.9% black, 0.5% American Indian and Alaska Native, 0.9% Asian, 73% Hispanic or Latinx and 1% other ethnicity) based on CalEnviroScreen 4.0; (2) pollution burden percentile is an average of 47% based on CalEnviroScreen 4.0; and (3) the City of Fillmore is not a DAC at the census tract level based on California Department of Water Resources (DWR) Disadvantaged Community Mapping Tool; however, the community in Fillmore is partially a DAC at the block group level.

Regarding water quality in the vicinity of the Census Tract Nos. 6111000302, 6111000303, and 6111000304, the Los Angeles Water Board found that (1) the water quality or aquifer risk in the majority of the City of Fillmore is medium in 2022; (2) the average drinking water contaminant percentile in Fillmore is 44, meaning it is higher than 44% of the census tracts in California; and (3) the census tract is not located in a priority basin where groundwater is overdrafted for potential saltwater intrusion.

67. Assembly Bill 2108 (AB 2108), “Water Policy: environmental justice: disadvantaged and tribal communities,” added sections 189.7 and 13149.2 to the California Water Code, effective January 1, 2023, and requires the Water Boards to conduct equitable, culturally relevant outreach when considering proposed discharges of waste that may have disproportionate impacts on water quality in disadvantaged communities or tribal communities. Also, for certain actions, the Water Board must adopt findings related to water quality impacts in disadvantaged or tribal communities and those related to environmental justice concerns.

The Fillmore WRP is not located in a disadvantaged community, but there are tribal communities in the area. This Order includes a time schedule for achieving an applicable water quality objective under CWC section 13263. The Los Angeles Water Board has satisfied the outreach requirements set forth in CWC section 189.7 by conducting outreach in affected tribal communities. Pursuant to CWC section 13149.2, the Los Angeles Water Board reviewed readily available information and information raised to the Los Angeles Water Board by interested persons concerning anticipated water quality impacts in disadvantaged or tribal communities resulting from adoption of this Order. The Los Angeles Water Board also considered environmental justice concerns within the Board’s authority and those raised by interested persons with regard to those impacts.

The Los Angeles Water Board acknowledges that there is an impaired water body, Santa Clara River Reaches 4A and 10, near the Fillmore WRP, but anticipates that the issuance of this Order will result in minimal water quality impacts and environmental justice concerns within the scope of the Los Angeles Water Board’s authority. Although this Order temporarily delays the effective date for meeting effluent limitations, the interim effluent limitations for chloride and boron are performance-based and established by a 95th percentile of the 12-month rolling

average from the 3rd quarter 2017 through the 2nd quarter 2023. The time schedule in Provision I.21 requires the City to complete certain tasks by specific deadlines so that it will comply with the new effluent limitations within the allotted time. This approach will allow the City to focus its resources on improving groundwater water quality, likely through system modifications to optimize the treatment and operation.

68. The Los Angeles Water Board notified the City, interested agencies and parties, and persons, on August 23, 2023, of the intent to revise the WDRs for this facility and has provided an opportunity to submit written comments by September 20, 2023. The public comment period was extended another 30 days to October 20, 2023 after discovery that the documents were inadvertently removed from the website after being posted for a few days.

On September 20, 2023, the City submitted comments on the tentative WDRs and WRRs. Los Angeles Water Board staff revised the tentative WDRs and WRRs to allow time for the City to come into compliance with the chloride and boron limitations and provided an opportunity for the public to submit written comments on the revised WDRs and WRRs by November 22, 2023.

On November 21, 2023, the City submitted written comments on the revised tentative WDRs and WRRs. Los Angeles Water Board staff have made changes to the revised tentative WDRs and WRRs based on the comments received.

69. The Los Angeles Water Board, in a public meeting, heard and considered all comments pertaining to the tentative waste discharge requirements.
70. Pursuant to CWC section 13320, any person affected by this action of the Los Angeles Water Board may petition the State Water Board to review the action in accordance with section 13320 of the CWC and CCR Title 23 section 2050. The State Water Board (P.O. Box 100, Sacramento, California 95812) must receive the petition within 30 days of the date this Order is adopted. The regulations regarding petitions may be found at [Water Quality Petitions | California State Water Resources Control Board](https://www.waterboards.ca.gov/public_notices/petitions/water_quality/index.html) (https://www.waterboards.ca.gov/public_notices/petitions/water_quality/index.html)

IT IS HEREBY ORDERED that the City of Fillmore shall be responsible for and shall comply with the following requirements in all operations and activities at the Fillmore WRP.

A. INFLUENT LIMITATIONS

1. Influent to the Fillmore WRP shall be limited to domestic, commercial, and industrial wastewater within Fillmore, RV waste, and septage. The Fillmore WRP shall not accept industrial wastewater specified in 40 CFR 403.5, *National pretreatment standards: Prohibited discharges*.

2. Daily average influent flowrate shall not exceed 1.8 mgd (or 2.4 mgd after expansion). Daily influent peak flowrate shall not exceed 5.0 mgd (or 6.0 mgd after expansion).
3. A pretreatment program shall be implemented consistent with the 40 CFR 403.8, *Pretreatment Program Requirements*, if nondomestic dischargers are found to introduce pollutants that may interfere with the treatment performance of the Fillmore WRP and impact the City's ability to meet the effluent limitations.

B. EFFLUENT AND RECYCLED WATER LIMITATIONS

1. Daily effluent average flowrate from the Fillmore WRP shall not exceed 1.8 mgd (or 2.4 mgd after expansion). Daily peak flowrate shall not exceed 5.0 mgd (or 6.0 mgd after expansion).
2. The pH of the effluent shall, at all times, be between 6.5 and 8.5 standard units.
3. Effluent quality shall not exceed the effluent limits listed in Table 19 below.

Table 19. Effluent Limitations

Constituents	Units	Daily Maximum	Monthly Average
BOD ₅ @20	mg/L	15	10
Total suspended solids	mg/L	15	10
Oil and grease	mg/L	15	10
Total dissolved solids	mg/L	2,000	None
Sulfate	mg/L	800	None
Total nitrogen	mg/L	10	None
Nitrate as nitrogen	mg/L	10	None
Nitrite as nitrogen	mg/L	1	None
MBAS	mg/L	0.5	None
Total coliform	MPN/100mL	None	1.1
Fecal coliform	MPN/100mL	None	1.1
Enterococcus	MPN/100mL	None	1.1

Table 19 notes:

- Total nitrogen consists of ammonia as nitrogen, nitrite as nitrogen, nitrate as nitrogen, and organic nitrogen.
- The unit of MPN/100mL denotes the most probable number per 100 milliliters.

4. Effluent that has been passed through a microfiltration shall not exceed any of the following:
 - a. 0.2 Nephelometric Turbidity Unit (NTU) more than 5 percent of the time within a 24-hour period; and
 - b. 0.5 NTU at any time.
5. The median concentration of total coliform bacteria measured in the disinfected effluent after the UV disinfection system and before the effluent storage tank shall not exceed 2.2 MPN/100mL utilizing the bacteriological results of the last seven days and the number of total coliform bacteria shall not exceed 23 MPN/100mL in more than one sample in any 30-day period. No sample shall exceed 240 MPN/100mL.
6. Discharge of chloride to disposal areas, consisting of percolation ponds, subsurface percolation zones, and percolation trench disposal system (as listed in Section D.1, *infra*) shall not exceed the 3-month rolling average effluent limitation. The 3-month rolling average effluent limitation is mass-based and determined as the average of daily discharge flowrates multiplied by the groundwater quality objective over three months. Daily mass loadings for chloride are calculated by the following equation and results are shown in Attachment E:

$$\text{Chloride} \left(\frac{\text{lbs}}{\text{day}} \right) = 8.34 \left(\frac{\text{L} \cdot \text{lbs}}{\text{mg} \cdot \text{MG}} \right) \times \text{total effluent flowrate (mgd)} \times 100 \left(\frac{\text{mg}}{\text{L}} \right)$$

7. Discharge of boron to disposal areas, consisting of percolation ponds, subsurface percolation zones, and percolation trench disposal system (as listed in Section D.1, *infra*) shall not exceed the 3-month rolling average effluent limitation. The 3-month rolling average effluent limitation is mass-based and determined as the average of daily discharge flowrates multiplied by the groundwater quality objective over three months. Daily mass loadings for boron are calculated by the following equation and results are shown in Attachment E:

$$\text{Boron} \left(\frac{\text{lbs}}{\text{day}} \right) = 8.34 \left(\frac{\text{L} \cdot \text{lbs}}{\text{mg} \cdot \text{MG}} \right) \times \text{total effluent flowrate (mgd)} \times 1 \left(\frac{\text{mg}}{\text{L}} \right)$$

8. Effluent shall not contain trace, toxic, and other constituents in concentrations exceeding the applicable MCLs for drinking water established by the DDW in sections 64431, 64442, 64443, 64444, 64449, and 64533 of CCR Title 22, Division 4, or subsequent revisions, or at levels that adversely affect the beneficial uses of receiving groundwater. The effluent shall, at all times, not exceed the following MCLs (Attachment A).

- a. Primary MCLs:
 - i. Inorganic chemicals in CCR Title 22, Division 4, Chapter 15, Section 64431, Table 64431-A (Attachment A-1);
 - ii. Radionuclides in CCR Title 22, Division 4, Chapter 15, Section 64442, Table 64442 (Attachment A-2) and Section 64443, Table 64443 (Attachment A-2); and
 - iii. Organic chemicals in CCR Title 22, Division 4, Chapter 15, Section 64444, Table 64444-A (Attachment A-3).
- b. Secondary MCLs specified in CCR Title 22, Division 4, Chapter 15, Section 64449, Table 64449-A (Attachment A-4).
- c. Disinfection byproducts specified in CCR Title 22, Division 4, Chapter 15.5 (Disinfectant Residuals, Disinfection Byproducts, and Disinfection Byproduct Precursors), Section 64533, Table 64533-A (Attachment A-5).

C. INTERIM EFFLUENT LIMITATIONS FOR CHLORIDE AND BORON

1. The City shall not exceed the interim effluent limitations, specified in Table 20 below.
2. The interim limitations shall not be applicable on and after November 1, 2026. Effluent limitations in Sections B.6 and B.7 shall apply effective November 1, 2026.

Table 20. Interim Effluent Limitations for Chloride and Boron

Constituents	Units	12-Month Rolling Average Concentrations
Chloride	mg/L	140
Boron	mg/L	1.6

D. GROUNDWATER LIMITATIONS

1. The discharge of treated wastewater from the Fillmore WRP shall not cause any exceedance of the groundwater limitations in Table 21 below. Compliance with the groundwater limitations will be determined at groundwater wells MW-1 through MW-10, specified in Finding 23.A, *supra*.

Table 21. Groundwater Limitations

Constituents	Units	Maximum
Total dissolved solids	mg/L	2,000
Sulfate	mg/L	800

Constituents	Units	Maximum
Chloride	mg/L	100
Boron	mg/L	1.0
Nitrate as nitrogen	mg/L	10
Nitrite as nitrogen	mg/L	1
Nitrate as nitrogen plus nitrite as nitrogen	mg/L	10
Total coliform	MPN/100mL	1.1
Fecal coliform	MPN/100mL	1.1
Enterococcus	MPN/100mL	1.1

2. Groundwater designated for use as domestic or municipal supply (MUN) shall not contain concentrations of chemical constituents and radionuclides in excess of the limits specified in Attachment A.
3. Groundwater shall not contain concentrations of chemical constituents in amounts that adversely affect any designated beneficial uses.
4. Groundwater shall not contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.

E. DISPOSAL AREA REQUIREMENTS

1. The authorized disposal areas are as follows:
 - Percolation ponds at the Fillmore WRP
 - Percolation ponds at the former Fillmore WWTP
 - Subsurface percolation zones at the Fillmore Middle School
 - Subsurface percolation zones at the Two Rivers Park
 - Percolation trench disposal system at the former Fillmore WWTP
2. The total disposal volumes to the disposal areas listed in Section E.1, *supra*, shall not exceed the calculated flowrate by equation 5 in Figure 12.
3. The City shall report the vertical separation distance measured in the quarterly monitoring report when discharging to the following disposal systems:
 - Subsurface percolation zones at the Fillmore Middle School,
 - Subsurface percolation zones at the Two Rivers Park, and
 - Percolation trench disposal system at the former Fillmore WWTP.

F. SPECIFICATIONS FOR PRODUCTION, OPERATIONS, AND USE OF RECYCLED WATER

1. The City is responsible for ensuring that appropriate ordinances are established to regulate production, operations, and use of recycled water and its facilities within the City of Fillmore and areas that overlie the Subbasin, consistent with this Order.
2. Recycled water shall not be used for direct human consumption or processing of foods or drink intended for human consumption.
3. The City is required to comply with all the conditions specified in the Title 22 Engineering Report and any amendments accepted by the DDW for recycled water applications prior to initiating new recycled water uses and/or use areas. For the recycled water use areas that were not included in the 2008 Title 22 Engineering Report and any future expansion shall be approved by the DDW before any implementation.
4. The delivery of recycled water to existing and new end-users that were not included in the 2008 Title 22 Engineering Report shall be subject to DDW acceptance and/or local public health agency.
5. No irrigation with disinfected tertiary recycled water shall take place within 50 feet of any domestic water supply well.
6. There shall be no storage or impoundment of disinfected tertiary recycled water within 100 feet of any domestic water supply well.
7. Recycled water irrigation rates shall not exceed agronomic rates of the plants. Special precautions must be taken to prevent clogging of spray nozzles and over-watering, and minimize surface runoff. Distribution pipelines shall be maintained to prevent leakage.
8. All aboveground irrigation appurtenances need to be marked appropriately.
9. Any incidental runoff shall be handled as follows:
 - A. The discharge of recycled water to surface water is prohibited.
 - B. Discharges of recycled water to surface waters may only occur where regulated under a separate National Pollutant Discharge Elimination System (NPDES) permit issued by the Los Angeles Water Board.

Incidental runoff is defined as unintended small amounts (volume) of runoff from recycled water use areas, such as unintended, minimal over-spray from sprinklers that escapes the recycled water use area. Irrigation system maintenance shall be consistent with the requirements found in the State Water Board's Recycled Water Policy.

10. Spray, mist, or runoff shall not enter dwellings, designated outdoor eating areas, or food handling facilities, and shall not reach to any drinking water fountain.
11. Recycled water shall not be used for irrigation during periods of rainfall and/or runoff.
12. Recycled water shall be retained in the designated area and shall not create surface runoff.
13. All recycled water use areas accessible to the public shall be posted with signs that are visible to the public, in a size no less than 4 inches high by 8 inches wide, and that include the following wording: "RECYCLED WATER – DO NOT DRINK" as shown in Figure 15. Each sign shall also display an international symbol similar to that shown in Figure 15. An alternative signage and wording must be approved by the Executive Officer of the Los Angeles Water Board.
14. No physical connection shall be made or allowed to exist between any recycled water piping and any piping conveying potable water, except as allowed under section 7604 of CCR Title 17.
15. The portions of the recycled water piping system that are in areas subject to access by the general public shall not include any hose bibs (a faucet or similar device to which a common garden hose can be readily attached). Only quick couplers that differ from those used on the potable water system shall be used on the portions of the recycled water piping system in areas subject to public access.
16. Recycled water use shall not result in earth movement in geologically unstable areas.
17. The City or its authorized agency shall develop the User Agreements and Ordinances with the potential non-potable users of recycled water. Copies of the User Agreements and Ordinances shall be provided to the Los Angeles Water Board and the DDW.
18. Site supervisors shall be appointed for the recycled water use areas, and their staff shall be trained on the hazards of working with recycled water and periodically retrained.
19. For each new recycled water use area, the City shall provide the Los Angeles Water Board and DDW with a description of the use area including, but not limited to: a description of the recycled water use (e.g., landscape, specific food crop, cooling tower, etc.); method of use (e.g., spray, flood, or drip); the location of domestic water supply facilities adjacent to the use areas; site containment measures; the party responsible for the distribution and use of the recycled water at the site; and identification of other governmental entities which may

have regulatory jurisdiction over the reuse site(s) such as State Food and Drug, State Licensing and Certification, County Health Department, etc. These Agencies shall also be provided with a copy of the approved Title 22 Engineering Report for review and comment.

G. GENERAL REQUIREMENTS

1. Standby or emergency power generation facilities with a sufficient capacity shall be provided for treated wastewater storage during rainfall or in the event of plant upsets or outages and at times when irrigation cannot be practiced.
2. Adequate facilities shall be provided to protect the Fillmore WRP, treatment system devices, sewer collection system, and disposal facilities from damage by storm flows and run-off or run-on generated by a 100-year return storm per 24-hour duration.
3. The treatment system, including the collection system that is a part of the treatment system and the disposal system, shall be maintained in such a manner that prevents sewage from surfacing or overflowing at any location.
4. A minimum of two feet of freeboard shall be maintained in the percolation ponds to ensure that direct rainfall will not cause overtopping.
5. No disposal areas with treated wastewater shall be located within 600 feet of any domestic water supply well unless all of the following conditions have been met:
 - a. A geological investigation demonstrates that an aquitard exists at the well between the uppermost aquifer being drawn from and the ground surface;
 - b. The well contains an annular seal that extends from the surface into the aquitard;
 - c. The well is housed to prevent any treated wastewater spray from coming into contact with the wellhead facilities;
 - d. The ground surface immediately around the wellhead is contoured to allow surface water to drain away from the well; and
 - e. The owner of the well approves of the elimination of the buffer zone requirement.
6. There shall be no storage or impoundment of treated wastewater within 600 feet of any domestic water supply well.
7. No disposal of sludge, waste, and treated wastewater shall take place within 600 feet of any reservoir or stream used as a source of domestic water.

8. Any wastes that do not meet the requirements in this Order shall be held in impervious containers and discharged at a legal point of disposal.
9. Percolation ponds shall be maintained to ensure that the percolation rate at the pond bottom shall not decrease over time.
10. The City shall notify well owners within a 500 feet radius from the boundary of percolation ponds within 10 days when there is any exceedance on the effluent limitations and/or groundwater limitations.
11. The City shall submit a Climate Change Effects Vulnerability Assessment and Management Plan (Climate Change Plan) no later than 12 months after the adoption of this permit or by **December 23, 2024**. Submittal of the Climate Change Plan is required pursuant to CWC section 13267. Under this provision, Los Angeles Water Board may require a person to submit technical or monitoring program reports upon request. The estimated costs of this report are approximately between \$40,000 and \$60,000. The burden, including costs of these reports, bears a reasonable relationship to the need for the reports and the benefits to be obtained from the reports, which include protection of public health and the environment. The Climate Change Plan is needed in order to assess and manage climate change related effects associated with the City's operation of the Fillmore WRP that may affect water quality.

The Climate Change Plan shall include an assessment of short- and long-term vulnerabilities of the facility(ies) and operations, as well as plans to vulnerabilities of collection systems, facilities, treatment systems, and discharge locations for predicted impacts in order to ensure that facility operations are not disrupted, compliance with permit conditions is achieved, and receiving waters, are not adversely impacted by discharges. Control measures shall include, but are not limited to, emergency procedures, contingency plans, alarm/notification systems, training, backup power and equipment, and the need for planned mitigations to ameliorate climate-induced impacts including, but not limited to, changing influent and receiving water quality and conditions, as well as the impact of rising sea level (where applicable) storm surges and back-to-back severe storms that are expected to become more frequent.

H. PROHIBITIONS

1. There shall be no wastewater overflows or discharge of partially treated wastewater from the treatment, storage, or disposal facilities to adjacent drainage ways, adjacent properties, or waters of the State at any time.
2. Wastes discharged shall not impact adverse tastes, odors, color, foaming or other objectionable characteristics to the receiving water.

3. There shall be no permanent onsite disposal of sludge. Sludge-drying activities are allowed, but only as an intermediate treatment prior to offsite disposal. Any offsite disposal of sewage or sludge shall be made only to a legal point of disposal. For the purposes of this Order, a legal disposal site is one for which requirements have been established by the Los Angeles Water Board or comparable regulatory entity and which is in full compliance therewith. Any sewage or sludge handling shall be in such a manner as to prevent its reaching surface waters or watercourses.
4. No part of the disposal system shall be closer than 100 feet to any water well.
5. Sewage odors from wastewater treatment shall not be detected beyond the property line.
6. Wastes discharged from the wastewater treatment plant shall at no time contain any substances in concentrations toxic to the human, animal, plant, or aquatic life.
7. The discharge of wastes shall not create a condition of pollution, contamination, or nuisance.
8. Nutrient-containing wastewater discharged to the percolation ponds shall not cause objectionable aquatic growth or degrade indigenous biota.
9. The direct or indirect discharge of any wastewater to surface waters or surface water drainage courses is prohibited without an NPDES permit.
10. The percolation ponds shall not contain floating materials, including solids, foams, or scum in concentrations that cause nuisance, adversely affect beneficial uses, or serve as a substrate for undesirable bacterial or algae growth or insect vectors.
11. The percolation ponds, drying beds, and the berms surrounding the ponds shall not contain plants, shrubs, or bushes that may damage the berms and the ponds.
12. Bypass (the intentional diversion of the waste stream from any portion of a treatment facility) is prohibited. The Los Angeles Water Board may take enforcement action against the City for bypass unless:
 - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage. (Severe property damage means substantial physical damage to property, damage to the treatment facilities that cause them to become inoperable, or substantial and permanent loss in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production);

- b. There were no feasible alternatives to bypass, such as the use of auxiliary treatment facilities, retention of untreated waste, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate backup equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that could occur during normal periods of equipment downtime or preventive maintenance; and
 - c. The City submitted a notice at least 48 hours in advance of the need for a bypass to the Los Angeles Water Board.
- 13. Any discharge of wastewater from the treatment system (including the wastewater collection system) at any point other than specifically described in this Order is prohibited and constitutes a violation of this Order.
- 14. No part of the treatment system and the percolation ponds shall extend to a depth below ground where wastes may deleteriously affect an aquifer that is usable for beneficial purposes. At all times, a minimum of 10 feet of vertical separation between the disposal system and the highest historical groundwater elevation or the water table must be maintained.
- 15. Wastes shall not be disposed of in geologically unstable areas to cause earth movement.
- 16. Adequate facilities shall be provided to divert surface and stormwater away from the treatment plant and disposal system and from areas where any potential pollutants are stored.
- 17. Discharge of waste classified as "hazardous," as defined in section 2521(a) of CCR Title 23, section 2510 et seq., is prohibited. Discharge of waste classified as "designated," as defined in CWC section 13173, in a manner that causes a violation of groundwater limitations is prohibited.

I. PROVISIONS

- 1. The City shall comply with all applicable requirements set forth in Chapter 4.5 (commencing with section 13290) of Division 7 of the California Water Code.
- 2. A copy of this Order shall be maintained at the wastewater treatment plant available at all times to operating personnel.
- 3. The City shall file with the Los Angeles Water Board self-monitoring reports on work performed according to the detailed specifications contained in the revised MRP No. CI-10753 attached hereto and incorporated herein by reference, as directed by the Executive Office. The results of any monitoring done more frequently than required at the location and/or times specified in the MRP shall be reported to the Los Angeles Water Board. The City shall comply with all of the provisions and requirements of the MRP.

4. Should effluent monitoring data indicate possible contamination of groundwater attributable to effluent from the Fillmore WRP, the City shall submit, within 90 days after discovery of the problem, mitigation measures that will be taken, or have been taken, to prevent further degradation that may result from the discharge(s).
5. The City shall participate in the implementation of the Watershed-wide Monitoring Program if the Executive Officer determines that a surface water monitoring program for the Santa Clara River is needed to fully evaluate the impact from City's effluent discharge on groundwater. The Los Angeles Water Board may require the City to participate with the Los Angeles Water Board, Santa Clara River Enhancement and Management Plan Steering Committee, and other stakeholders in the development and implementation of a watershed-wide monitoring program.
6. In accordance with CWC section 13260(c), the City shall file with the Los Angeles Water Board a report of any material change or proposed change in the character, location, or volume of the discharge.
7. The City shall operate and maintain its wastewater collection, treatment, and disposal facilities in a manner to ensure that all facilities are adequately staffed, supervised, financed, operated, maintained, repaired, and upgraded as necessary to provide adequate and reliable transport, treatment, and disposal of all wastewaters from both existing and planned future wastewater sources under the City's responsibilities. Anyone employed in the operations of the wastewater treatment plant must be certified pursuant to CWC sections 13625 through 13633 and in accordance with CCR Title 23, section 3680.

8. Collection System Requirements

The City must properly enroll under the applicable statewide general order for sanitary sewer collection systems (i.e., General WDRs for SSS or any successor permit) and comply with the requirements specified in the statewide general order.

The City's collection system is part of the system that is subject to Order WQ 2022-0103-DWQ. As such, the City must properly operate and maintain its collection system (40 CFR part 122.41(e)). The City must report any non-compliance (40 CFR part 122.41(l)(6) and (7)) and mitigate any discharge from the collection system in violation of this Order (40 CFR part 122.41(d)).

9. Spill Reporting Requirements

- a. **Initial Notification** – Although State and Los Angeles Water Board staff do not have duties as first responders, this requirement is an appropriate mechanism to ensure that the agencies that do have first responder duties are notified in a timely manner in order to protect public health and

beneficial uses. For certain spills, overflows, and bypasses, the City shall make notifications as required below:

- i. In accordance with the requirements of Health and Safety Code section 5411.5, the City shall provide notification to the local health officer or the director of environmental health with jurisdiction over the affected water body of any unauthorized release of sewage or other waste that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but no later than two hours after becoming aware of the release.
- ii. In accordance with the requirements of CWC section 13271, the City shall provide notification to the California Office of Emergency Services (Cal OES) of the release of reportable quantities of hazardous substances or sewage that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but not later than two hours after becoming aware of the release. CCR Title 23, section 2250, established 1,000 gallons or more as a reportable quantity of sewage. The phone number for reporting these releases to the Cal OES is (800) 852-7550. The City shall also include public outreach in their emergency communications protocols, which may include media updates, social media postings, and community notices.
- iii. The City shall notify the Los Angeles Water Board of any unauthorized release of sewage from the Fillmore WRP that causes, or probably will cause, a discharge to a water of the state as soon as possible, but not later than two hours after becoming aware of the release. This initial notification does not need to be made if the City has notified Cal OES and the local health officer or the director of environmental health with jurisdiction over the affected waterbody. The phone number for reporting these releases of sewage to the Los Angeles Water Board is (213) 620-2150. The phone numbers for after-hours and weekend reporting of releases of sewage to the Los Angeles Water Board are (213) 305-2284 and (213) 305-2253.

At a minimum, the following information shall be provided to the Los Angeles Water Board:

- The location, date, and time of the release;
- The water body that may be impacted by the discharge;
- An estimate of the amount of sewage or other waste released and the amount that reached the receiving water at the time of notification;
- If ongoing, the estimated flowrate of the release at the time of the notification;
- The name, organization, phone number, and email address of the reporting representative; and

- A certification that Cal OES and the local health officer or directors of environmental health with jurisdiction over the possibly affected water bodies have been notified of the discharge.

- b. **Monitoring** – For spills, overflows, and bypasses reported under Section H of this Order, the City shall monitor as required below:

To define the geographical extent of the spill's impact, the City shall obtain grab samples (if feasible, accessible, and safe) for all spills, overflows, or bypasses of any volume that reach any waters of the State (including surface and ground waters). The City shall analyze the samples for total and fecal coliform, *Escherichia coli* (*E. coli*), enterococcus, and relevant pollutants of concern, upstream and downstream of the point of entry of the spill (if feasible, accessible and safe). This monitoring shall be done on a daily basis from time the spill is known until the results of two consecutive sets of bacteriological monitoring indicate the return to the background level or the County Department of Public Health authorizes cessation of monitoring.

- c. **Reporting** – The initial notification required under Section H.13 above shall be followed:

- i. As soon as possible, but not later than 24 hours after becoming aware of an unauthorized discharge of sewage or other waste from its wastewater treatment plant to a water of the state, the City shall submit a statement to the Los Angeles Water Board staff via email. If the discharge is 1,000 gallons or more, this statement shall certify that Cal OES has been notified of the discharge in accordance with CWC section 13271. The statement shall also certify that the local health officer or director of environmental health with jurisdiction over the affected water bodies has been notified of the discharge in accordance with Health and Safety Code section 5411.5. The statement shall also include, at a minimum, the following information:
- Agency, Order No., and MRP No.;
 - The location, date, and time of the discharge;
 - The water body that received the discharge;
 - A description of the level of treatment of the sewage or other waste discharged;
 - An initial estimate of the amount of sewage or other waste released and the amount that reached the impacted water body;
 - The Cal OES control number and the date and time that notification of the incident was provided to Cal OES; and
 - The name of the local health officer or director of an environmental health representative notified (if contacted directly); the date and time of notification; and the method of notification (e.g., phone, fax, email).

- ii. A written preliminary report shall be submitted to the Los Angeles Water Board within five working days after disclosure of the incident via the State Water Board GeoTracker database under Global ID WDR100056013. The final written report shall be included in the next quarterly monitoring report submitted to the GeoTracker database above. The written report shall document the information required in paragraph Section I.9.d below, monitoring results, and any other information required in provisions of the Standard Provisions document, including corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences.
 - iii. The City shall include a certification in the annual summary report (due according to the schedule in the accompanying MRP) that states that the sewer system emergency equipment, including alarm systems, backup pumps, standby power generators, and other critical emergency pump station components were maintained and tested in accordance with the City's preventive maintenance plan. Any deviations from or modifications to the Plan shall be discussed.
- d. **Records** – The City shall prepare and maintain a record of all spills, overflows, or bypasses of raw or partially treated sewage from its collection system or the Fillmore WRP. This record shall be made available to the Los Angeles Water Board upon request, and a spill summary shall be included in the annual report, as required by MRP No. CI-10753. The record shall contain:
 - i. The date and time of each spill, overflow, or bypass;
 - ii. The location of each spill, overflow, or bypass;
 - iii. The estimated volume of each spill, overflow, or bypass, including gross volume, amount recovered, and amount not recovered, monitoring results as required by Section H.9.b above;
 - iv. The cause of each spill, overflow, or bypass;
 - v. Whether each spill, overflow, or bypass entered receiving water and, if so, the name of the water body and whether it entered via storm drains or other man-made conveyances;
 - vi. Any corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences; and
 - vii. The mandatory information included in SSO online reporting for finalizing and certifying the SSO report for each spill, overflow, or bypass under the General WDRs for SSS.
 - viii. Emergency communications, such as media updates, social media postings, and community notices for purposes of public outreach as required by Section I.9.a.ii.
- e. **Activities Coordination** – The Los Angeles Water Board expects that the City will coordinate their compliance activities for consistency and efficiency with other entities that have responsibilities to implement: (1) this Order and (2) the General WDRs for SSS.

- f. **Consistency with the General WDRs for SSS** – The Los Angeles Water Board recognizes that there may be some overlap between the provisions in this Order and requirements in the General WDRs for SSS. To encourage efficiency, the Los Angeles Water Board will accept the documentation prepared by the City under the General WDRs for SSS for compliance purposes.
10. The City shall submit an updated Operations and Maintenance Manual (O & M Manual) for the Fillmore WRP and its disposal facilities to the Los Angeles Water Board within 90 days if there is any change to the original O & M Manual. The City shall maintain the O & M Manual in a usable condition and available for reference and use by all applicable personnel. The City shall regularly review and revise or update, as necessary, the O & M Manual in order for the document(s) to remain useful and relevant to current equipment and operation practices. Reviews shall be conducted annually, and revisions or updates shall be completed as necessary and submitted to the Los Angeles Water Board.
11. The City shall take all reasonable steps to minimize or prevent any discharge that has a reasonable likelihood of adversely affecting human health or the environment.
12. The City shall ensure that the capacity of the disposal area is adequate for the discharge and that adequate steps are taken to accommodate system failures and/or to deal with the loss of the soil assimilative capacity.
13. The City shall allow the treatment and disposal systems to be inspected annually during the life of this Order by an inspector retained by the City. The inspector shall specify the conditions of the treatment system and the disposal system. The inspector should also assess the capacities of the percolation ponds system and waste sludge drying beds system.
14. The City shall file a written report with the Los Angeles Water Board within 90 days after the average dry-weather flow for any month equals or exceeds 90 percent of the design capacity of the waste treatment and/or disposal facilities. The report shall detail provisions to cope with flows in excess of 90 percent of the design capacity.
15. For any violation of requirements in this Order, the City shall notify the Los Angeles Water Board within 24 hours of knowledge of the violation either by telephone or electronic mail. The notification shall be followed by a written report within one week. The City in the next monitoring report shall also confirm this information. In addition, the report shall include the reasons for the violations or adverse conditions, the steps being taken to correct the problem (including dates thereof), and the steps being taken to prevent a recurrence.
16. This Order does not relieve the City from the responsibility to obtain other necessary local, state, and federal permits to construct facilities necessary for

compliance with this Order; nor does this Order prevent the imposition of additional standards, requirements, or conditions by any other regulatory agency.

17. After notice and opportunity for a hearing, this Order may be terminated or modified for causes including, but not limited to:
 - a. Violation of any term or condition contained in this Order;
 - b. Obtaining this Order by misrepresentation or failure to disclose all relevant facts; or
 - c. A change in any condition or the discovery of any information that requires either a temporary or permanent reduction or elimination of the authorized discharge.
18. The City shall furnish, within a reasonable time, any information the Los Angeles Water Board may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order. The City shall also furnish to the Los Angeles Water Board, upon request, copies of records required to be kept by this Order.
19. This Order includes the attached *Standard Provisions Applicable to Waste Discharge Requirements* (Attachment B), which are incorporated wherein by reference. If there is any conflict between the provisions stated herein and the *Standard Provisions Applicable to Waste Discharge Requirements*, the provisions stated herein will prevail.
20. The City shall allow the Los Angeles Water Board or an authorized representative, upon the presentation of credentials and other documents as may be required by law, to:
 - a. Enter the City premises where a regulated facility or activity is located or conducted, or where records must be kept under the conditions of this Order;
 - b. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order;
 - c. Inspect, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order; and
 - d. Sample or monitor at reasonable times, for the purposes of assuring compliance with this Order, or as otherwise authorized by the California Water Code, any substances or parameters at any locations.

21. The City shall comply with the following requirements and schedules. Compliance may require system modifications to optimize the treatment and operation:
 - a. Submit a report on alternative potable water supply sources with concentration levels below groundwater quality objectives no later than October 31, 2024.
 - b. No later than June 30, 2025, submit a feasibility report for compliance with the boron and chloride effluent limitations in Provisions B.6 and B.7 by the November 1, 2026 effluent limitation compliance date. Method of compliance may include, but not be limited to, increasing recycled water usage rates, expansion of recycled water use areas, upgrade of the treatment system for salt removal and chloride and boron reduction. The feasibility study report shall include the selected method of compliance and implementation schedule.
 - c. Implement the selected method of compliance no later than June 30, 2026.
 - d. Achieve full compliance with effluent limitations no later than November 1, 2026.

J. COMPLIANCE DETERMINATION

1. Compliance with the effluent limitations contained in Section B of this Order will be determined as specified below:

- a. General:

Compliance determinations shall be based on available analyses for the time interval associated with the effluent limitations. Where only one sample analysis is available in a specified time interval (e.g., monthly or weekly average), that sample shall serve to characterize the discharge for the entire interval. If quarterly sample results show non-compliance with the average monthly limit and that sample result is used for compliance determinations for each month of the quarter, then three separate violations of the average monthly limit shall be deemed to have occurred.

- b. Monthly Average Effluent Limitation:

If the average of daily discharges over a calendar month exceeds the monthly average effluent limitation for a given parameter in Table 19, this will represent a single violation, though the City will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month). If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the monthly average effluent limitation, the City will be

considered out of compliance for that calendar month. However, the City will only be considered out of compliance for days when the discharge occurs. For any one calendar month during which no sample is taken due to no daily discharge, no effluent compliance determination can be made for that calendar month.

c. Average Weekly Effluent Limitation

If the average of daily discharges over a calendar week exceeds the average weekly effluent limitation for a given parameter, an alleged violation will be flagged and the City will be considered out of compliance for each day of that week for that parameter, resulting in 7 days of non-compliance. The average of daily discharges over the calendar week that exceeds the average weekly effluent limitation for a parameter will be considered out of compliance for that week only. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the average weekly effluent limitation, the City will be considered out of compliance for that calendar week. For any one calendar week during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar week with respect to the average weekly effluent limitation.

A calendar week will begin on Sunday and end on Saturday. Partial calendar weeks at the end of calendar month will be carried forward to the next month in order to calculate and report a consecutive seven-day average value on Saturday.

d. Daily Maximum Effluent Limitation:

If a daily discharge exceeds the daily maximum effluent limitation for a given parameter in Table 19, the City will be considered out of compliance for that parameter for that 1 day only within the reporting period. For any 1 day during which no sample is taken, no effluent compliance determination can be made for that day.

K. WORK PLAN TO PREVENT FURTHER GROUNDWATER DEGRADATION

The City shall submit a work plan to prevent further groundwater degradation when the City projects that 10 percent of the Subbasin assimilative capacity for chloride and boron will be used up within three years.

L. REOPENER

The Los Angeles Water Board will review this Order periodically and will revise requirements when necessary. This Order was developed based on currently available technical information and applicable water quality laws, regulations, policies, and plans to assure compliance with them. If applicable laws and

regulations change, including but not limited to, establishment of total maximum daily loads, or once new information is obtained that will change the overall discharge and its potential to impact waters of the state, it may be appropriate to reopen this Order. Additionally, the Los Angeles Water Board at its discretion may reissue this Order to change the implementation schedule specified in Provision I.21 based on the findings from the feasibility study report.

M. TERMINATION

Except for enforcement purposes, Time Schedule Order No. R4-2006-0050 adopted by the Los Angeles Water Board on May 11, 2006, and Waste Discharge Requirements Order No. R4-2017-0186 adopted on October 5, 2017, and amended on September 13, 2018, and October 8, 2020, are hereby terminated.

The Los Angeles Water Board's termination of prior waste discharge requirements and/or monitoring and reporting requirements do not extinguish any violations that may have occurred during the time those requirements were in effect. The Los Angeles Water Board reserves the right to take enforcement actions to address violations of prior prohibitions, limitations, specifications, requirements, or provisions of rescinded requirements as allowed by law.

N. EFFECTIVE DATE

This Order becomes effective immediately upon its adoption.

I, Susana Arredondo, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Los Angeles Region, on December 21, 2023.

Jenny Newman
Digitally signed by Jenny Newman
Date: 2024.01.10 12:11:44 -08'00'

for Susana Arredondo
Executive Officer

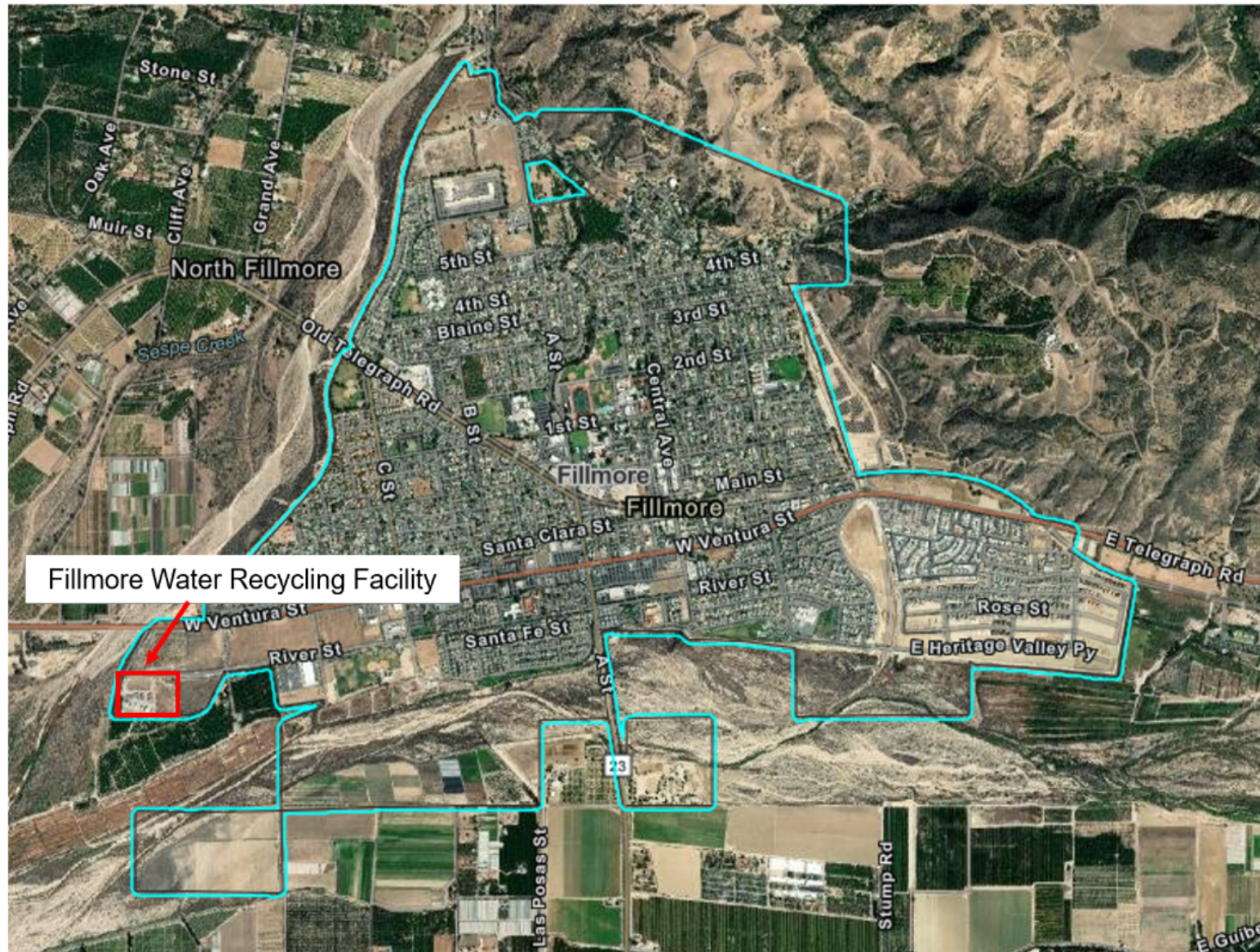


Figure 1. Vicinity of the Fillmore Water Recycling Facility and Boundary of the City of Fillmore.

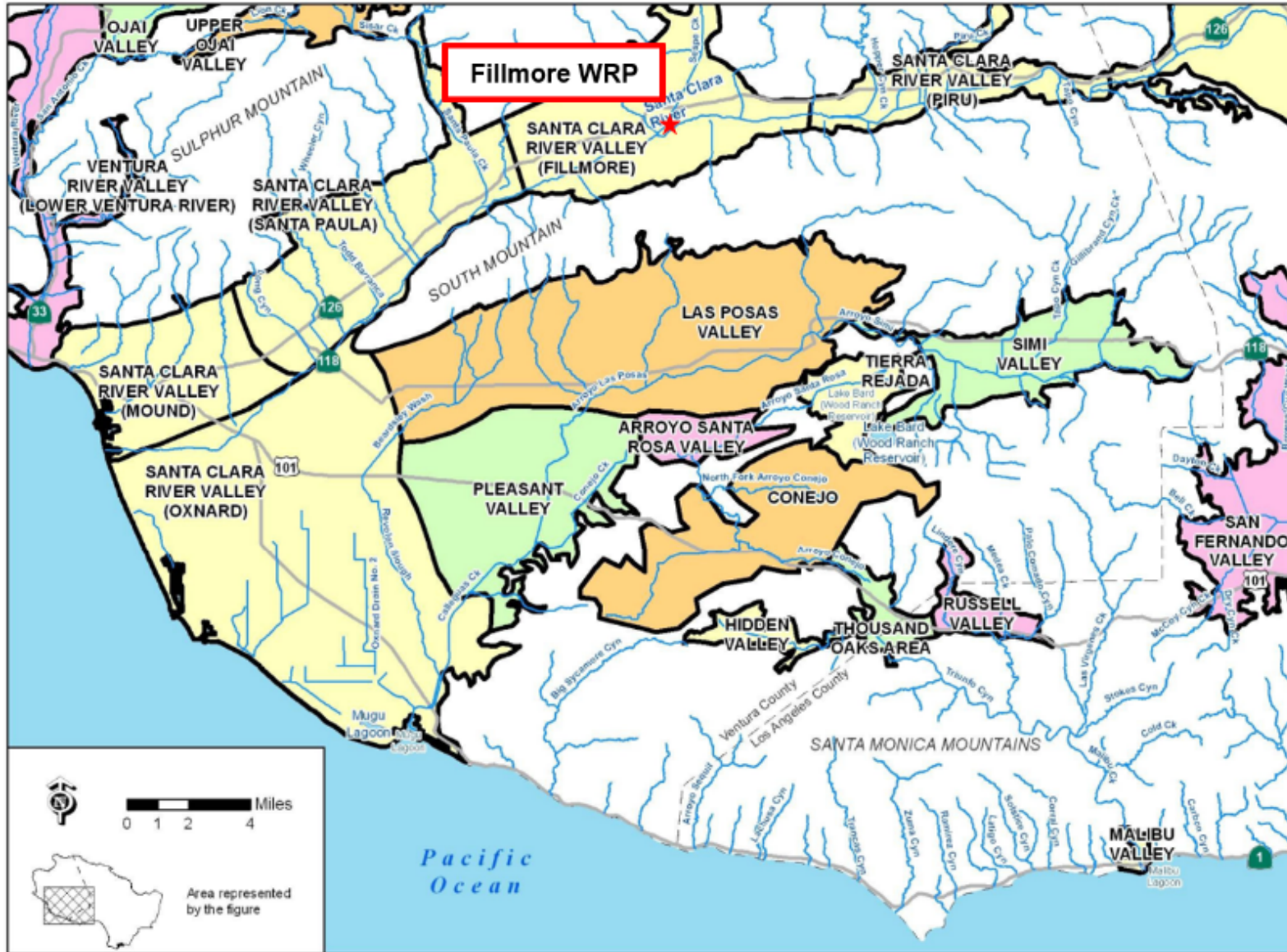


Figure 2. Fillmore Groundwater Basin.

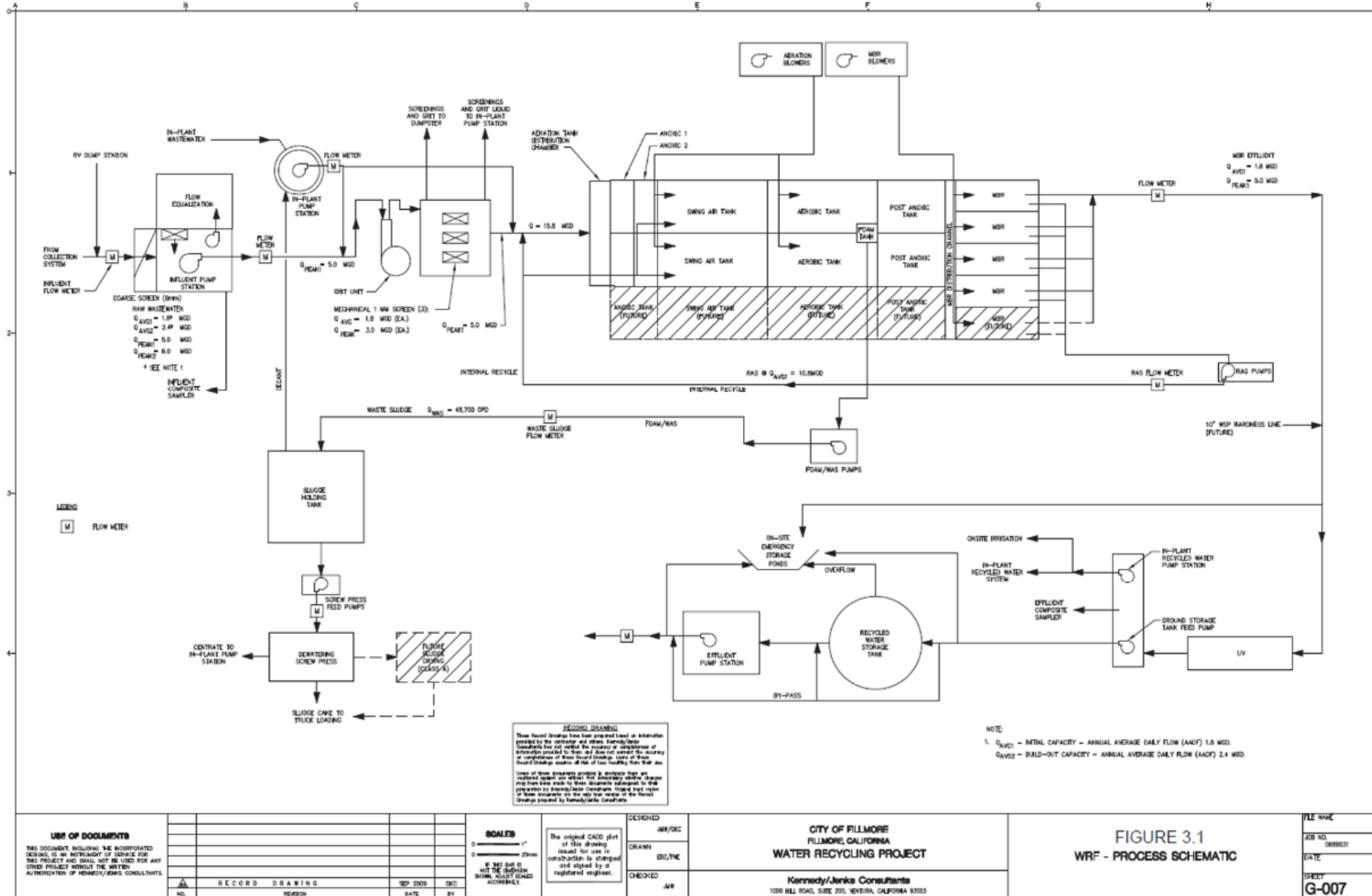


Figure 3. Fillmore Water Recycling Plant Process Flow Diagram.

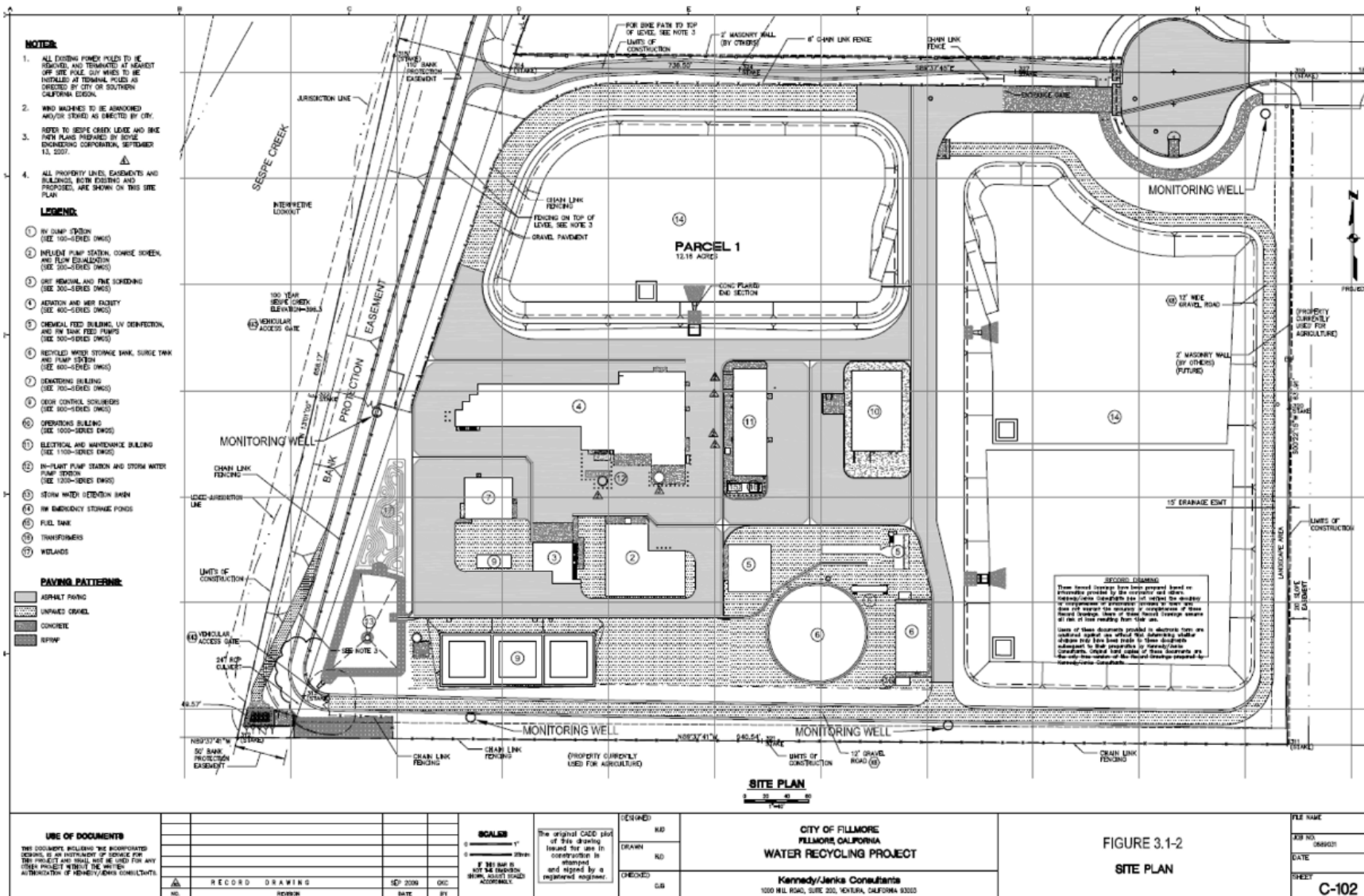


Figure 4. Site Plan of the Fillmore Water Recycling Plant.

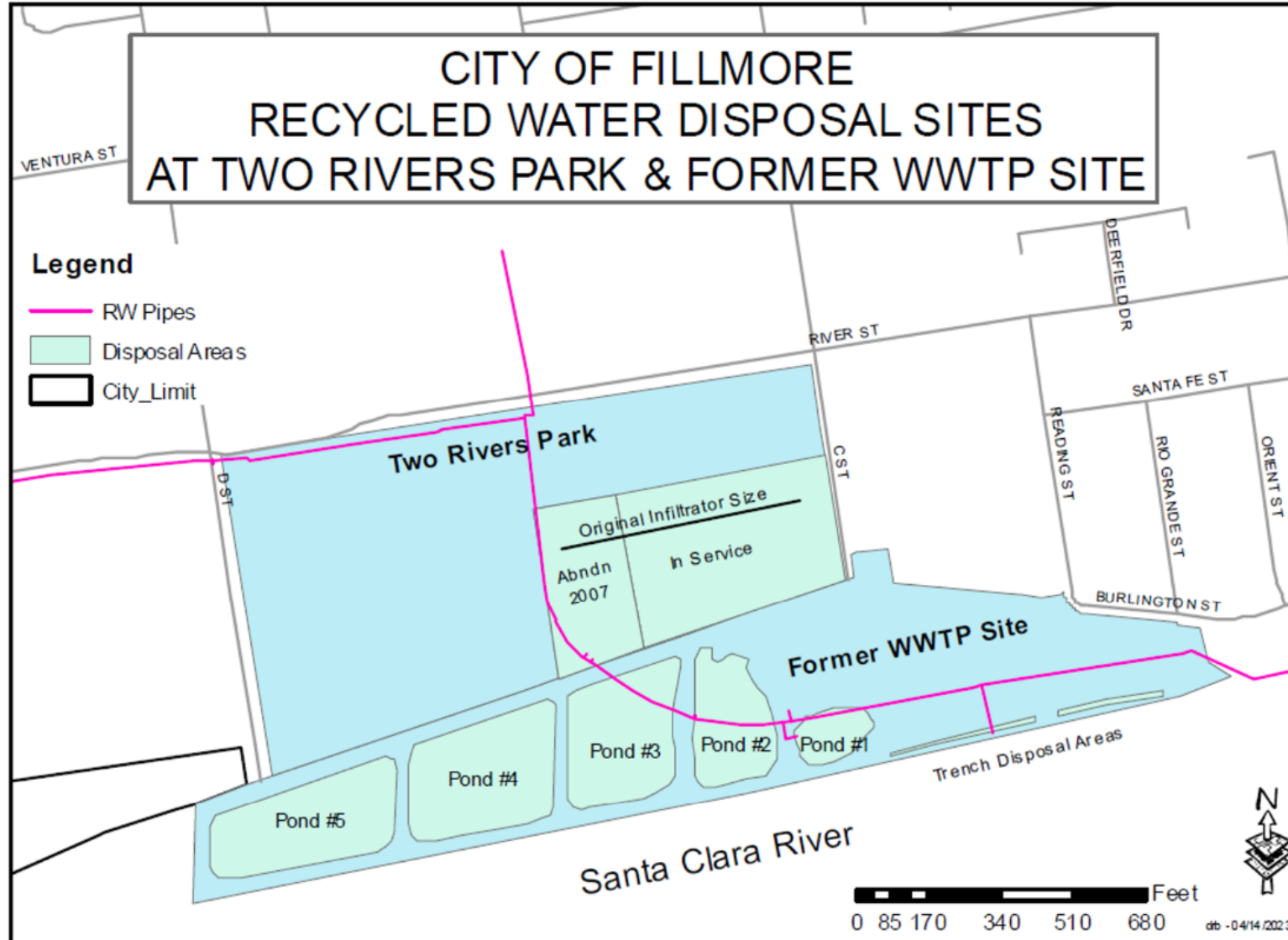


Figure 5. Percolation Ponds and Trench Disposal Systems at the Former Fillmore Wastewater Treatment Plant and Subsurface Percolation System at Two Rivers Park.

Attachment 04-4

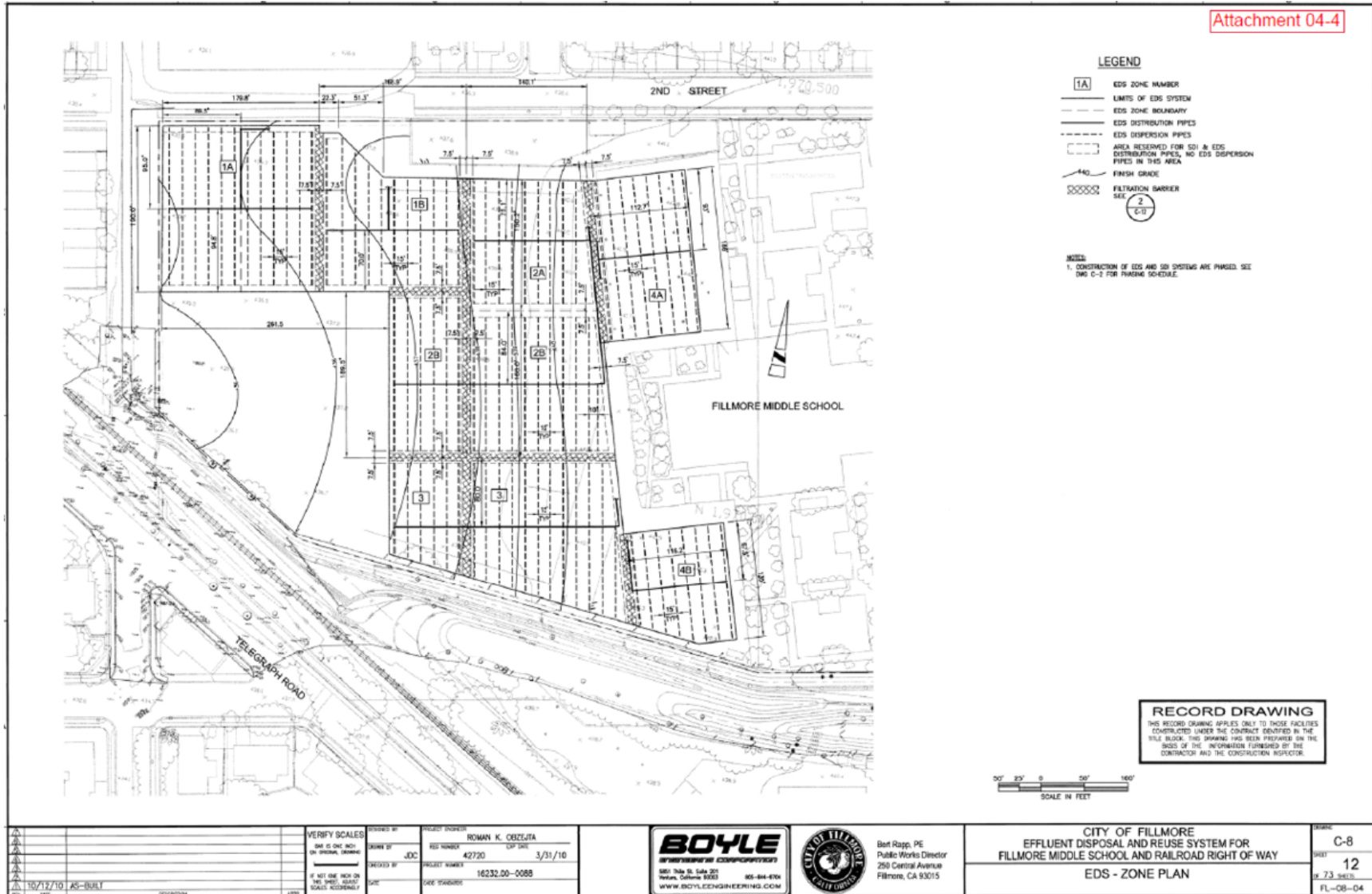


Figure 6. Subsurface Percolation System at Fillmore Middle School.

Attachment 02-1

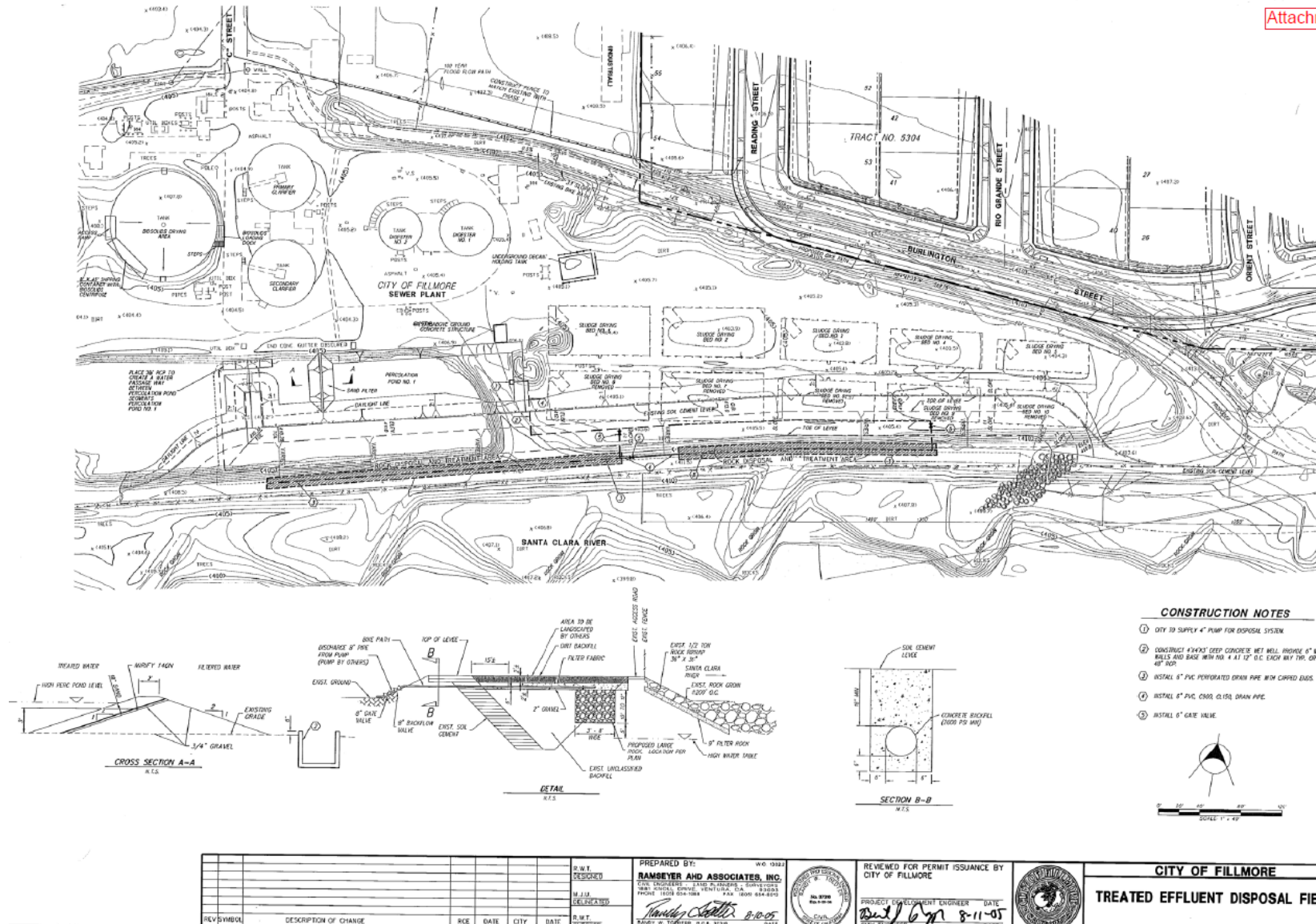


Figure 7. Trench Disposal System at the Former Fillmore Wastewater Treatment Plant.

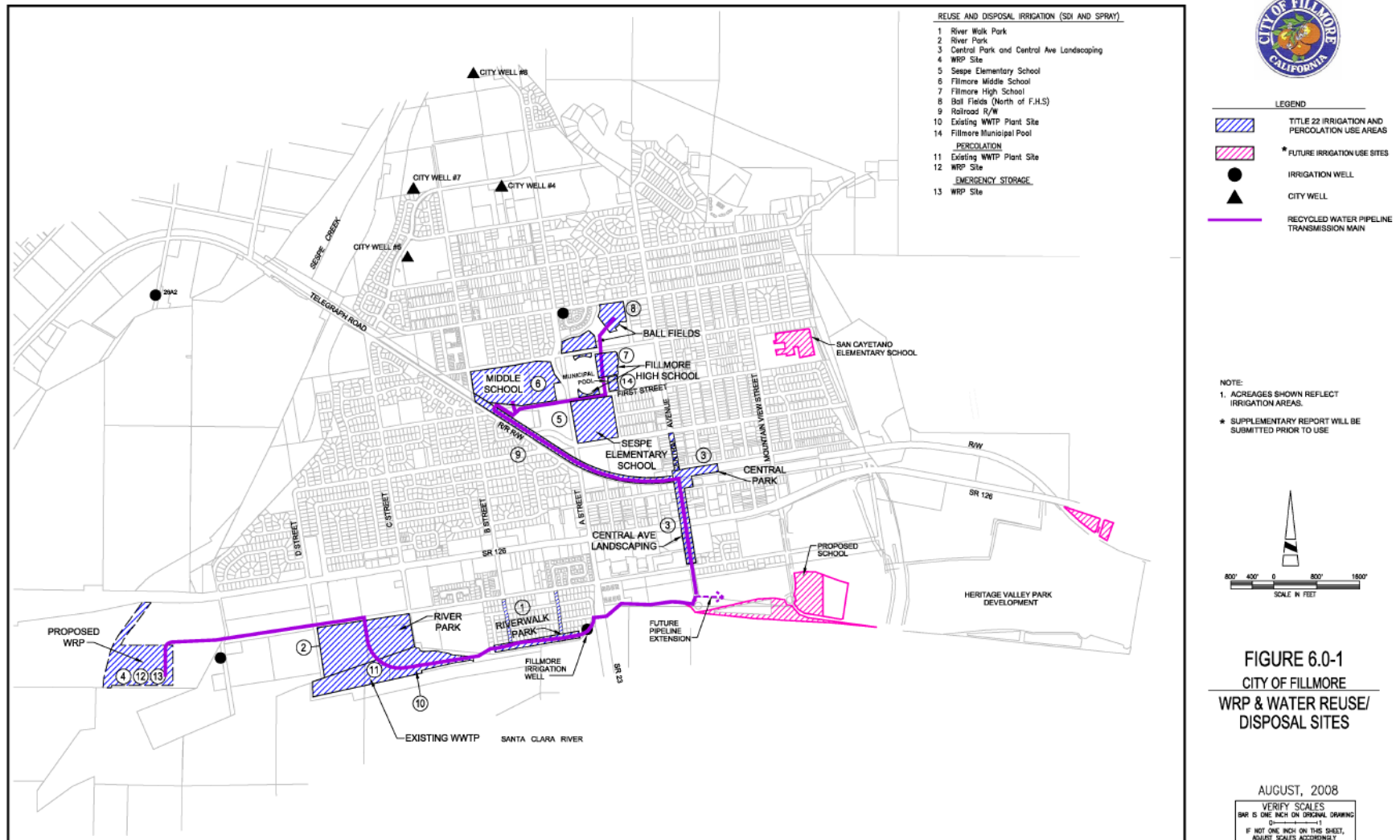


Figure 8. Recycled Water Use Areas and Recycled Water Distribution Pipelines Specified in the 2008 Title 22 Engineering Report.

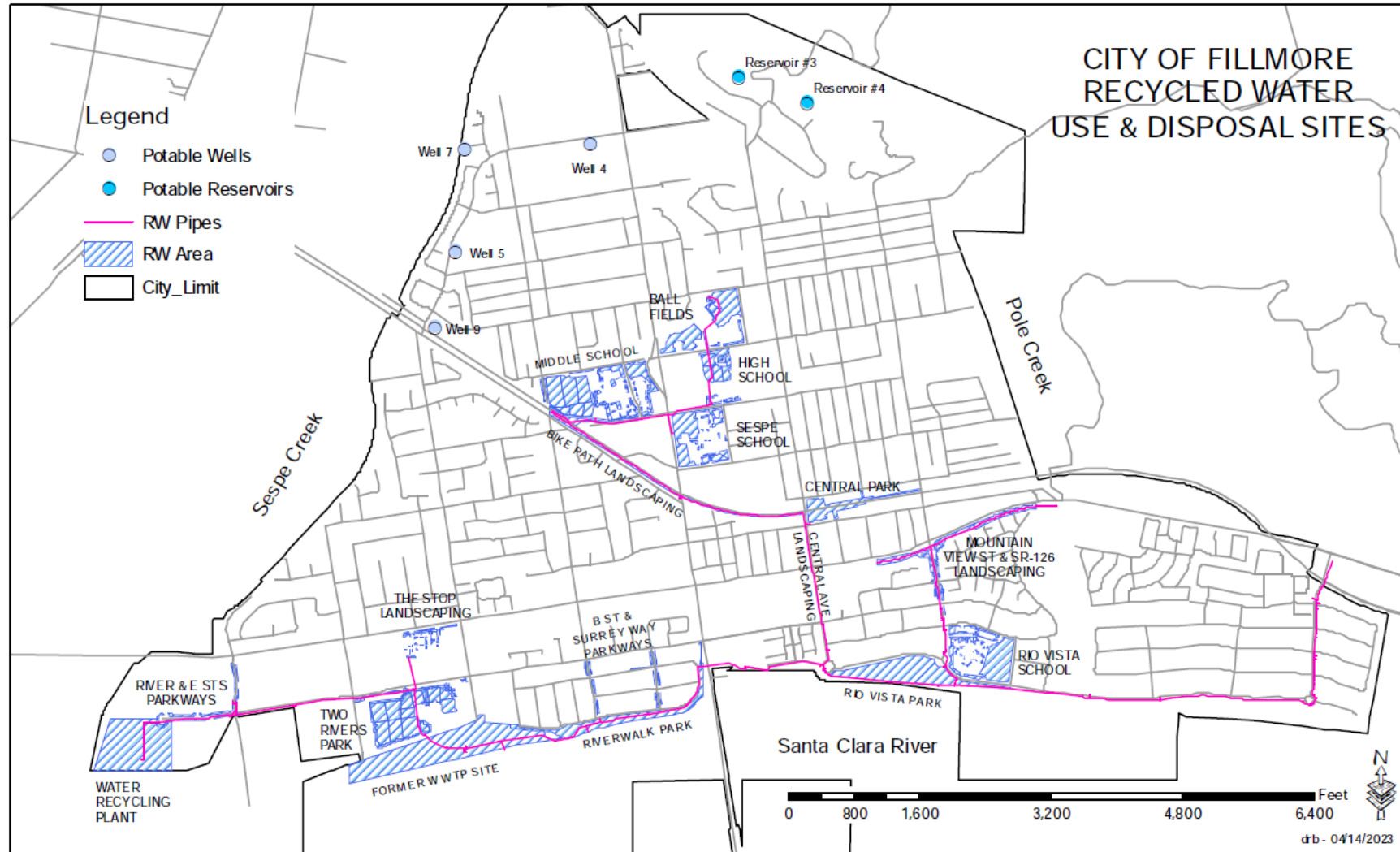


Figure 9. Current Recycled Water Use Areas and Recycled Water Distribution Pipelines.



Legend

-  Groundwater Monitoring Wells Installed Prior to 2009
-  Groundwater Monitoring Wells Installed March, 2009
-  Ventura County Key Well

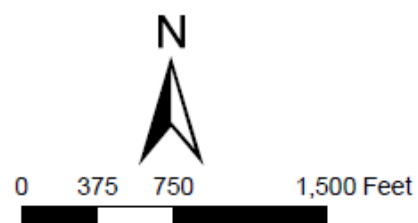


Figure 10. Layout of Groundwater Monitoring Wells.

City of Fillmore Water Recycling Plant
Calculations of Maximum Effluent Discharges of Boron & Chloride

I. Theoretical Max Allowable Discharge of Boron & Chloride per Day at Max Flow

From WDR's			Maximum Daily Discharge	
Max Average Daily Flow (MGD)	Max Boron Concentration (mg/L)	Max Chloride Concentration (mg/L)	Boron (Lbs/Day)	Chloride (Lbs/Day)
1.8	1.00	100	15.0	1499

II. Average Daily Discharge of Boron & Chloride at Current Average Flows & Concentrations

Curr. Average Daily Flow (MGD)	Curr. Avg Boron Concentration (mg/L)	Curr. Avg Chloride Concentration (mg/L)	Average Daily Discharge	
			Boron (Lbs/Day)	Chloride (Lbs/Day)
0.88	1.16	114	8.5	835
=cells to input				

III. Maximum Allowable Discharge of Boron & Chloride at Current Average Flows

Curr. Average Daily Flow (MGD)	Max WDR Boron Concentration (mg/L)	Max WDR Chloride Concentration (mg/L)	Max Allowable Daily Discharge	
			Boron (Lbs/Day)	Chloride (Lbs/Day)
0.88	1.00	100	7.3	733

Amount of Boron & Chloride to NOT be Discharged to the Percolation Ponds (lbs/Day) to achieve limits	Boron	Chloride
	1.2	102.6
Gallons of Effluent to be Recycled to other sites and therefore Not Discharge this Amount of Boron and Chloride at Current Concentrations in Gallons per Day	121,379	108,070

Figure 11. City's Proposed Chloride and Boron Mass Loadings.

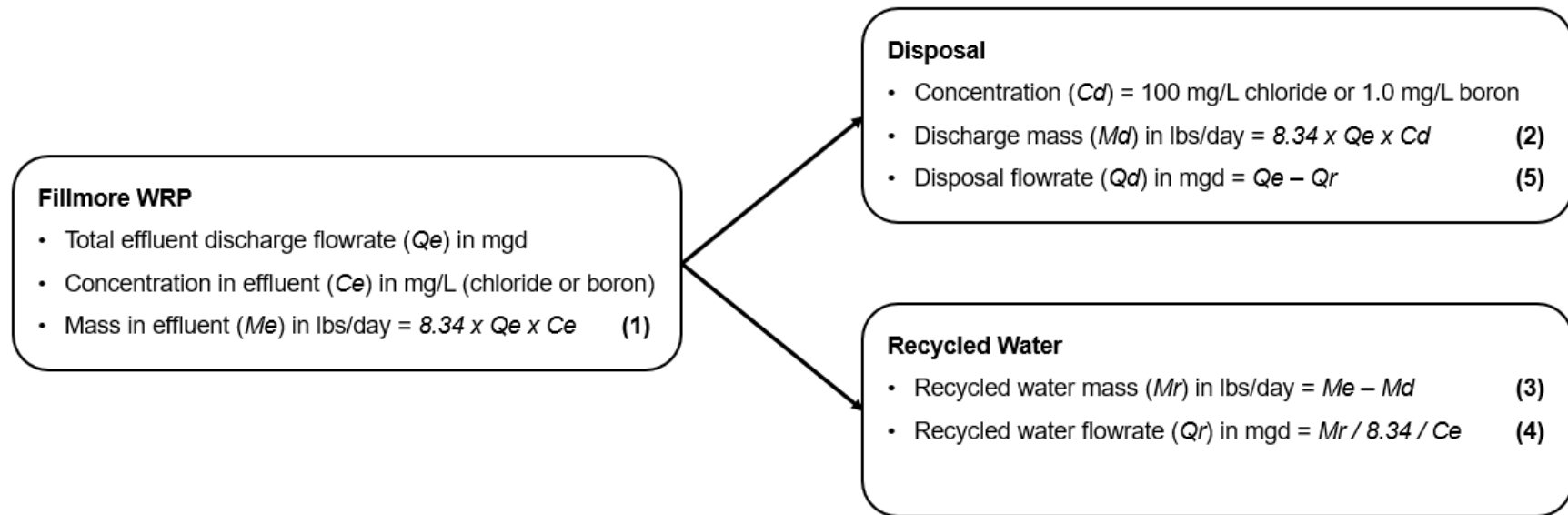


Figure 12. Chloride and Boron Mass Loadings Management Decision Diagram.

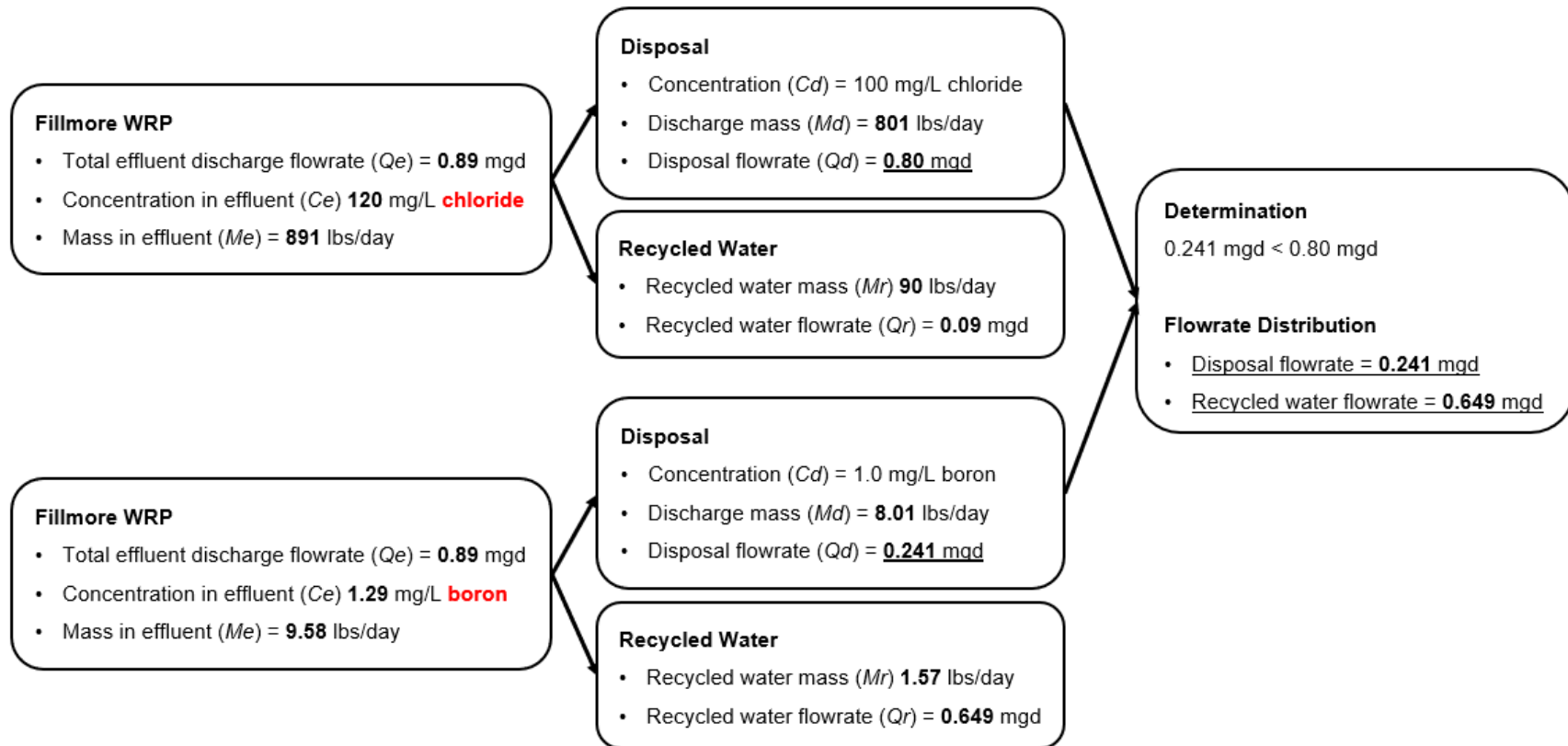


Figure 13. Example for Determining Effluent Discharge Flow Distribution.

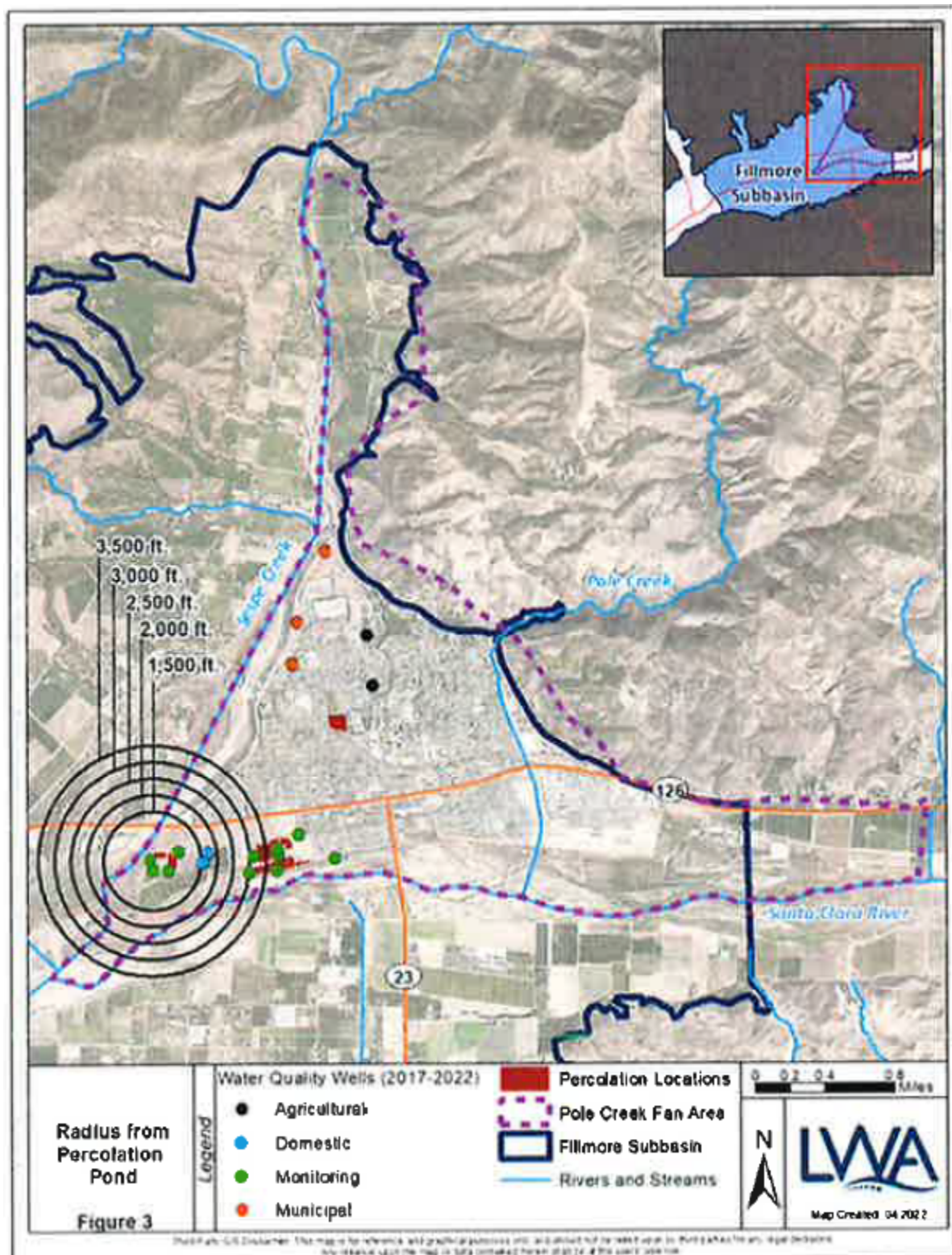


Figure 14. Groundwater Impact Analysis at Radii of 1,500 feet, 2,000 feet, 2,500 feet, 3,000 feet, and 3,500 feet from the Percolation Ponds.



Figure 15. Exhibit of “Recycled Water – Do Not Drink” Signage.